Global synchronization of inductively defined graph classes

a local-to-global perspective

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phd defense talk

Social Networks Lab

ETH Zürich

Outline

Synchronization and Kuramoto model

Global synchronization & Related problems & Existing works

- Preliminary: standard reduction

Preliminary

- Preliminary: proof strategy in previous works
- Our central tool: phasor geometry

Main results

- Results one by one

Summary & Outlook

Synchronization Phenomenon & Kuramoto model

Synchronization Phenomenon & Kuramoto model

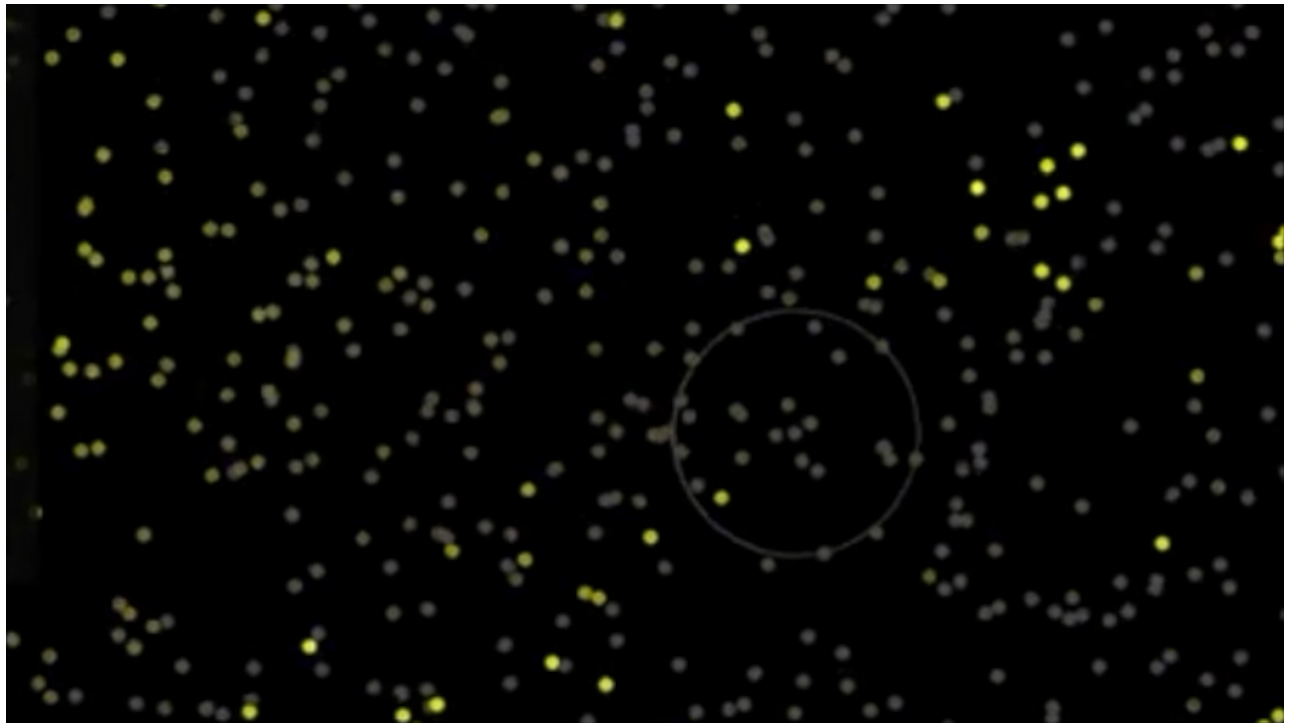
- Synchronization is everywhere

Synchronization Phenomenon & Kuramoto model

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- Different $\xrightarrow{\text{over time}}$ Same

Synchronization Phenomenon & Kuramoto model

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- Different $\xrightarrow{\text{over time}}$ Same
- Typical examples:

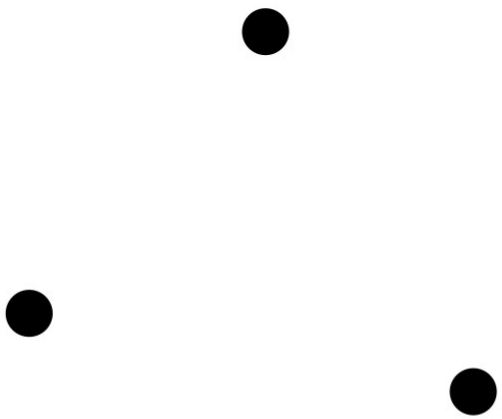


Flashing fireflies..

visualize-it.github.io

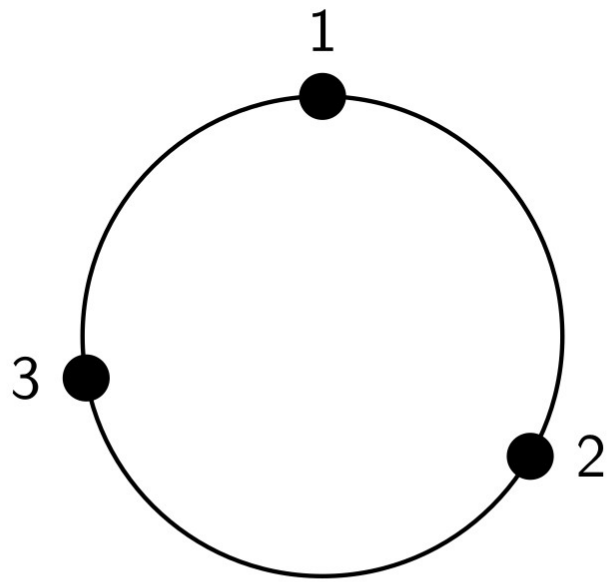
Synchronization Phenomenon & Kuramoto model

$$\forall i \in [n], \quad \frac{d\theta_i}{dt} = \sum_{j=1}^{n} A_{ij} \sin(\theta_j - \theta_i)$$



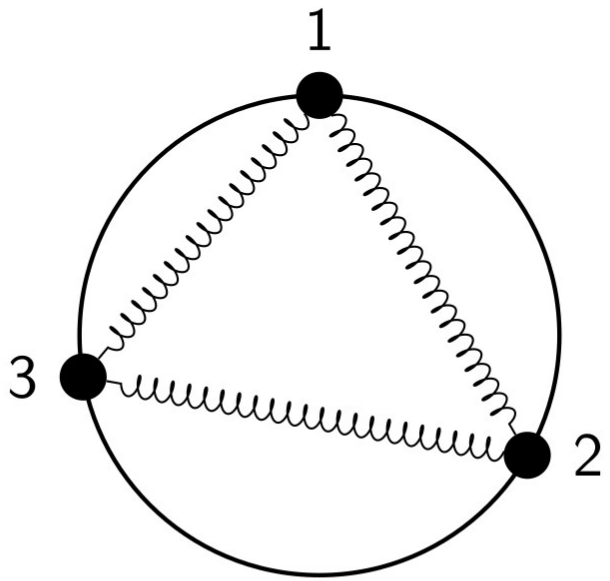
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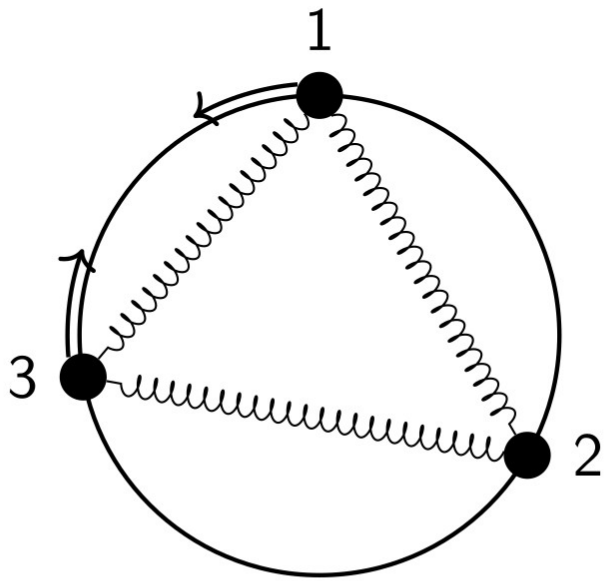
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The spring *drags* its two endpoints synchronize.

Global synchronization problem & Related problems & Existing works



For which graphs, does this process always drive all nodes converge to a single point, regardless of their initial placement?

Global synchronization problem & Related problems & Existing works



Global sync graphs

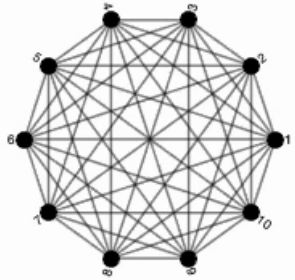
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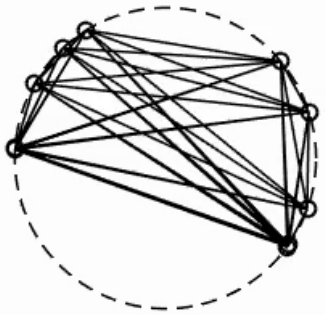
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Examples.





✔ Complete graph

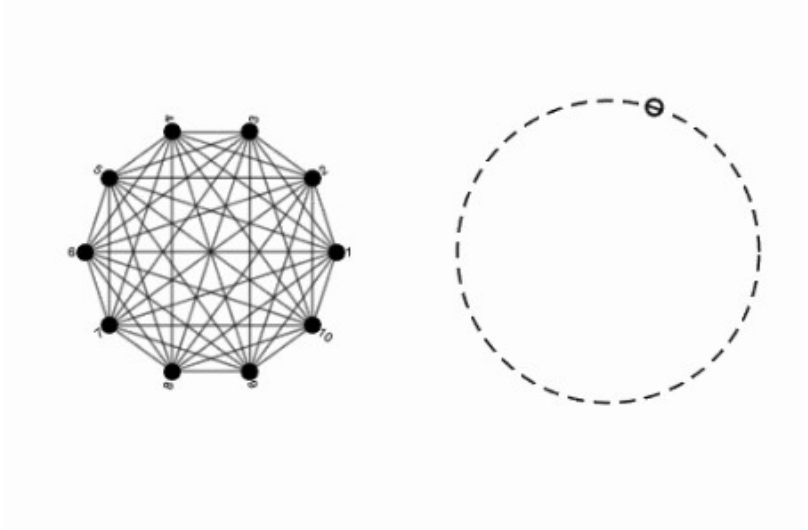
✘ Ring graph $n \geq 5$

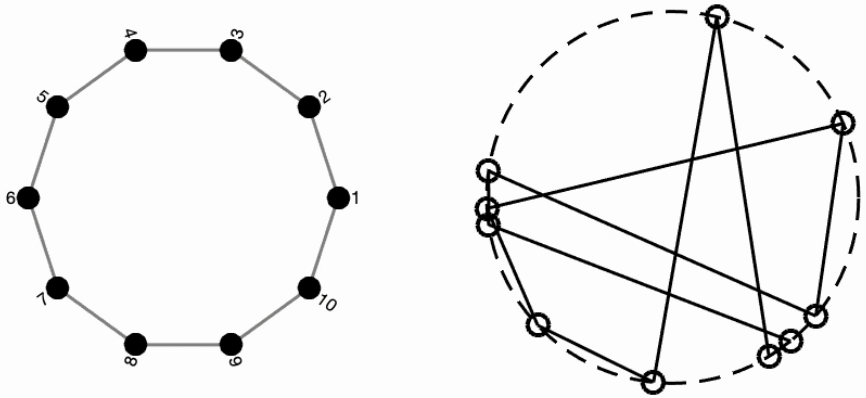
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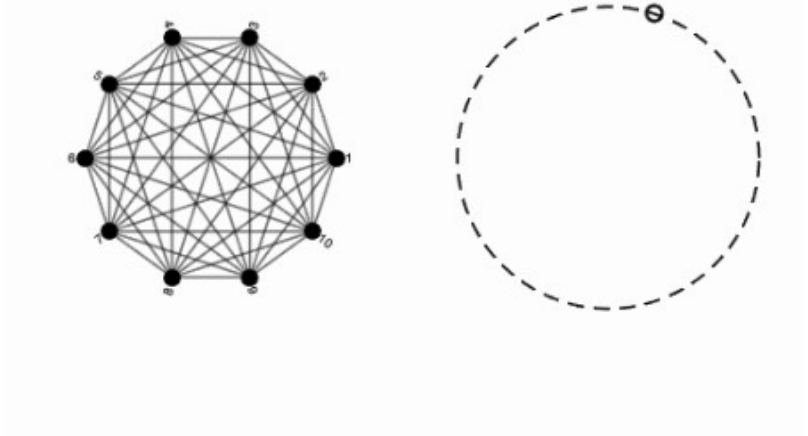
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Global synchronization problem & Related problems & Existing works

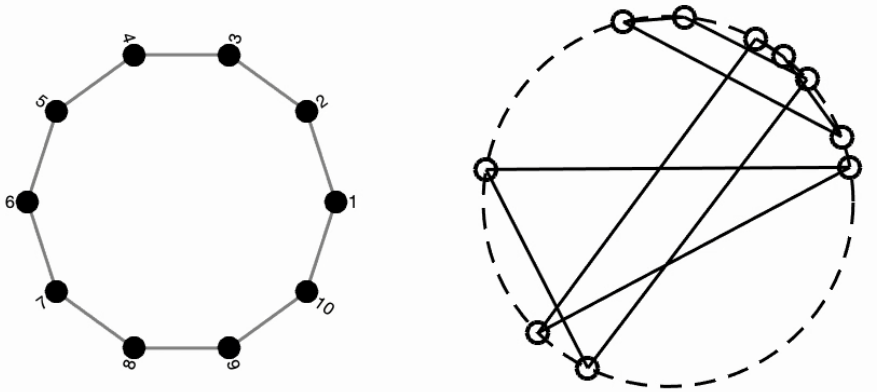
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Global synchronization & Related problems & Existing works

17th-century

modern

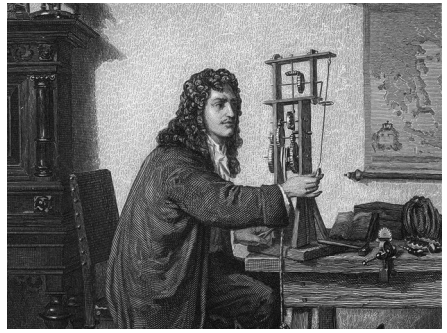
Huygens' observation on synchronized pendulum clocks

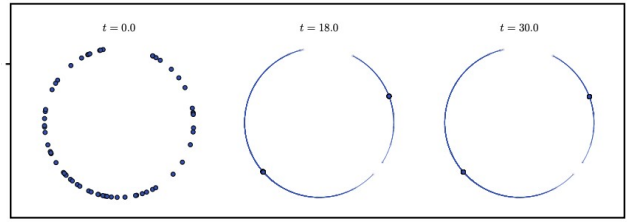
$\Rightarrow$

- benign nonconvex optimization

♀ WU

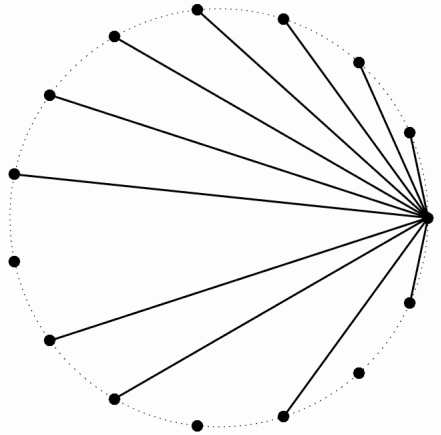
- clustering phenomena in transformer neural networks[1]





[1] 2023 Geshkovski, Letrouit, Polyanskiy, Rigollet: A mathematical perspective on Transformers

Global synchronization & Related problems & Existing works



Graph is incredibly dense → Global synchrony

every node has $\geq \mu_c(n-1)$ neighbours,

$\mu_c \in [0.6875, 0.75]$

[1] Taylor[2012], Ling & Xu & Bandeira[2018], Lu & Steinerberger[2021]

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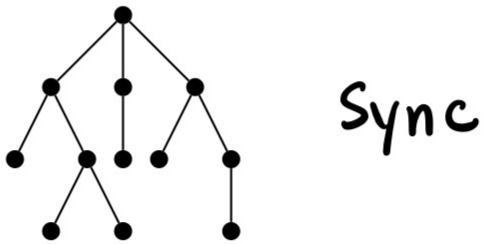
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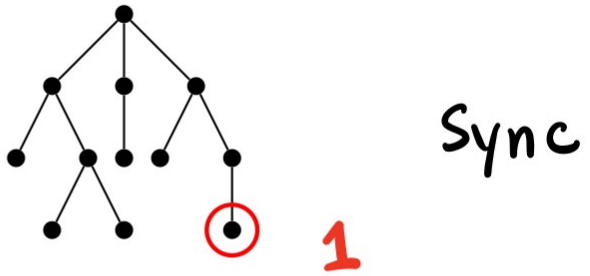
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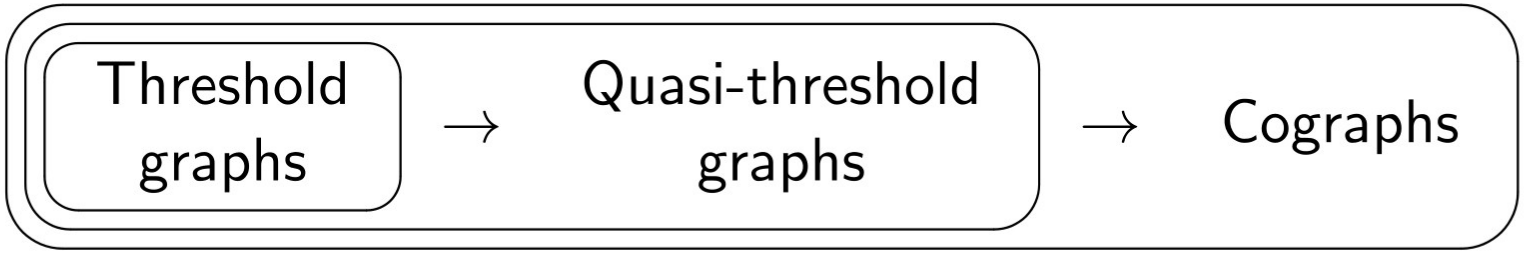
Except for incredibly dense graphs

There must exists other mechanisms driving global synchrony

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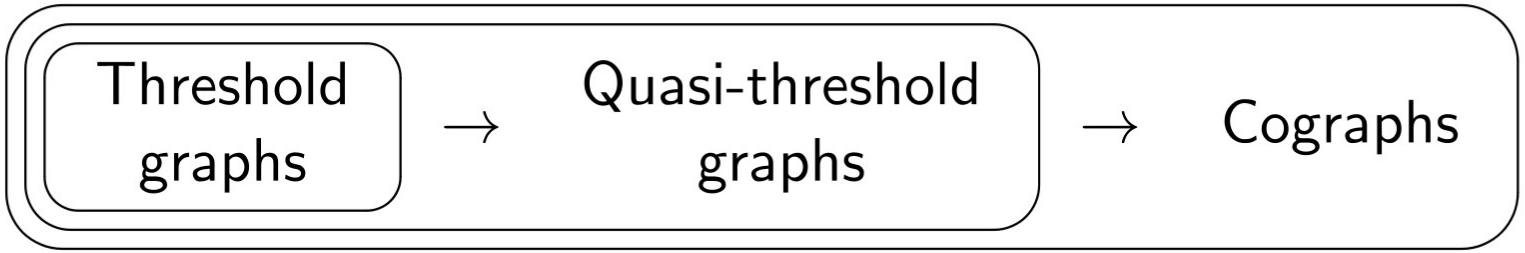
Thesis result

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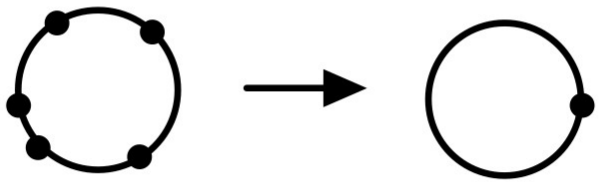


- New mechanism: Global sync can arises from local to global propagation.

some preparations

Preliminary: boil down the problem

G global sync $\xleftarrow{\text{implies}}$

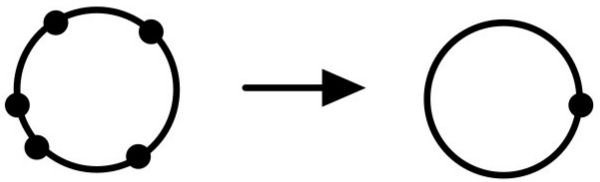


Every 2nd-order stationary point $\boldsymbol{\theta}$ of energy function $E(\boldsymbol{\theta})$ of Kuramoto model is sync state, that is $\theta_1 = \cdots = \theta_n$.

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why?

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why? geometry shapes dynamics.

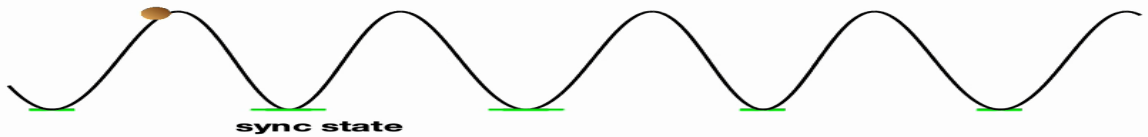
$$\frac{d\boldsymbol{\theta}}{dt} = -\nabla E(\boldsymbol{\theta}), \quad \text{where} \quad E(\boldsymbol{\theta}) = -\sum_{i=1}^{n}\sum_{j=1}^{n} A_{ij}\cos(\theta_i - \theta_j), \text{ up to a const.}$$

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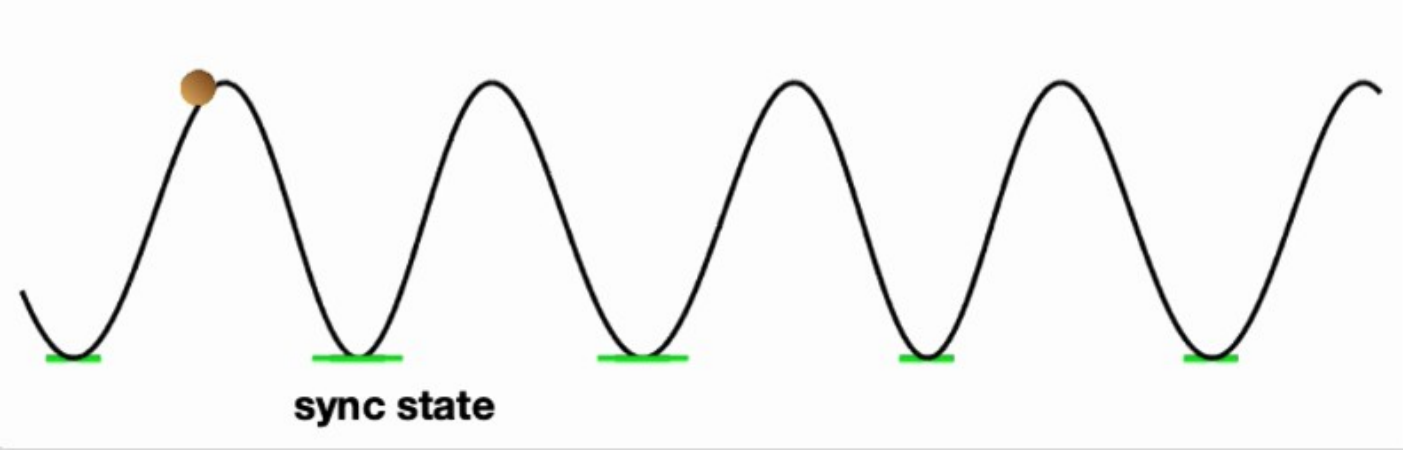


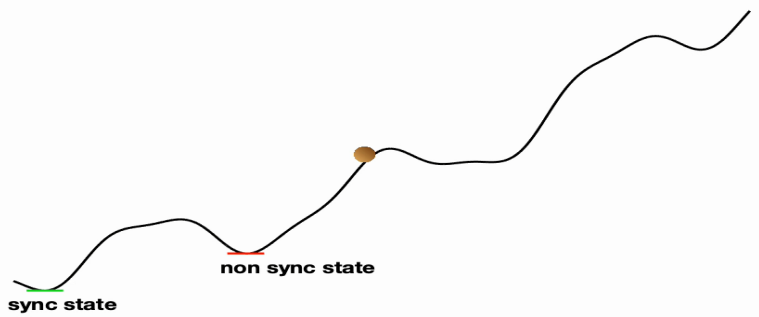
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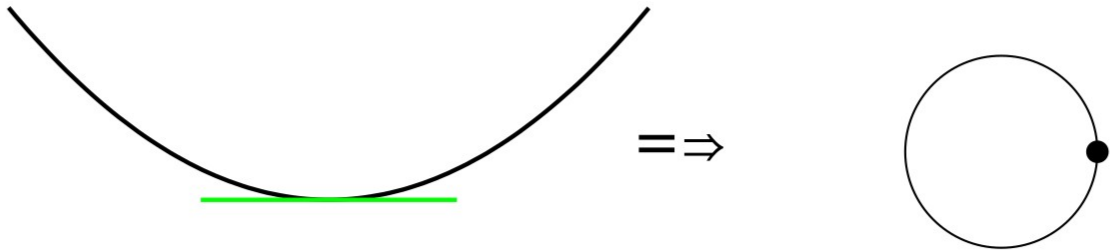
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Conceptually,

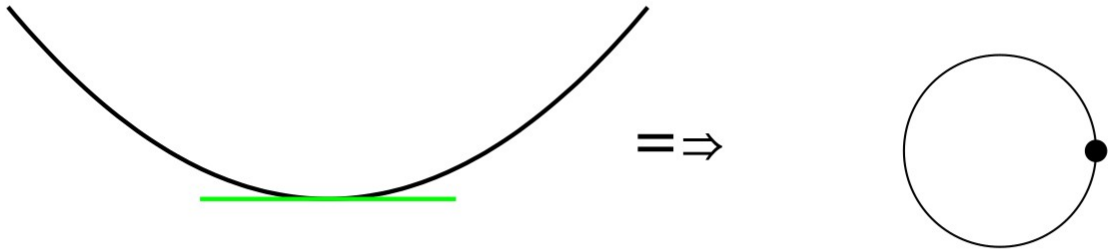


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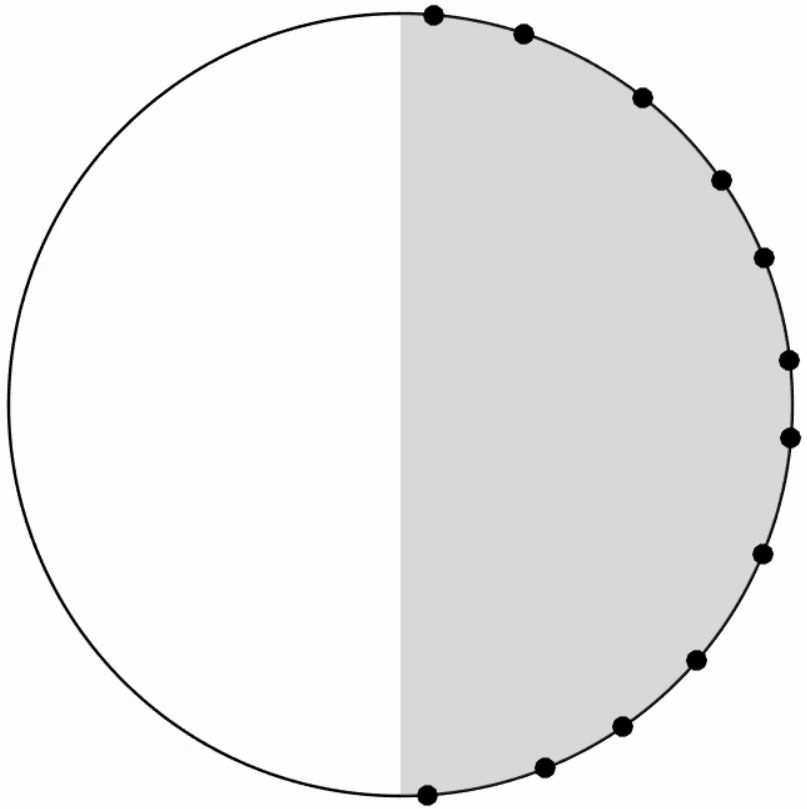
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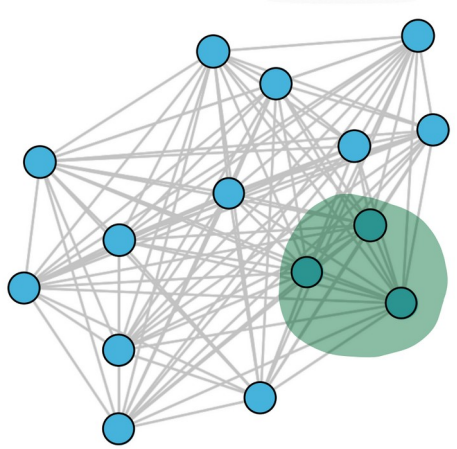
Preliminary: proof strategy in previous works

any second order stationary point $\boldsymbol{\theta} \in$



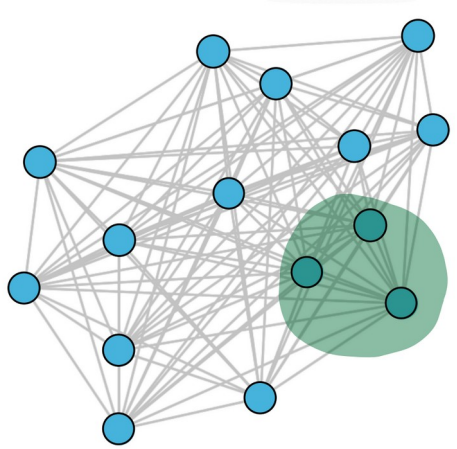
Our central tool: phasor geometry

- At a second-order point θ, we aim to directly understand the angles of small subsets of nodes through their structural relations in the graph.



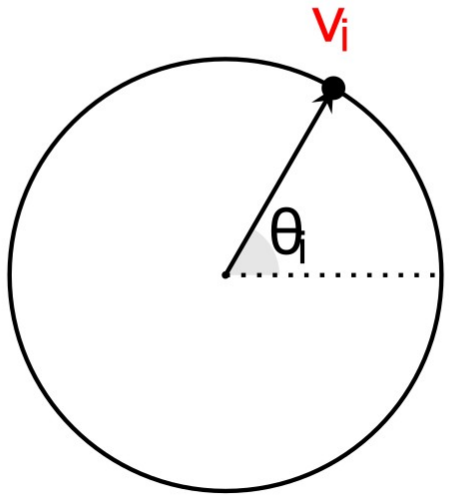
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- Denote v_i as Phasor of node i

Phaser

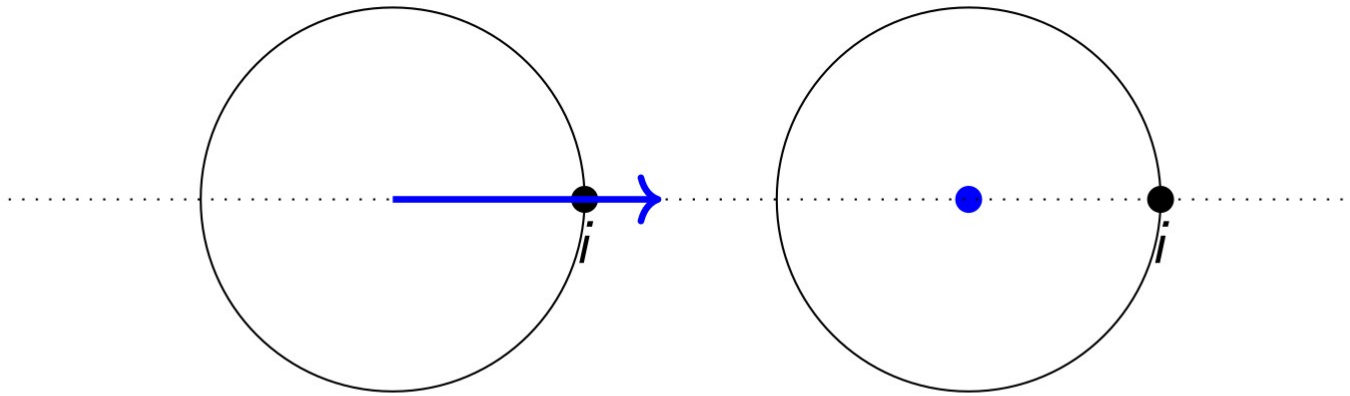


phase

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Lemma (single node at 2nd-order point)

If $\boldsymbol{\theta}$ is 2nd order SP, then $\sum_{j \in N(i)} \boldsymbol{v}_j = \mu_i \boldsymbol{v}_i, \quad \mu_i \geq 0.$

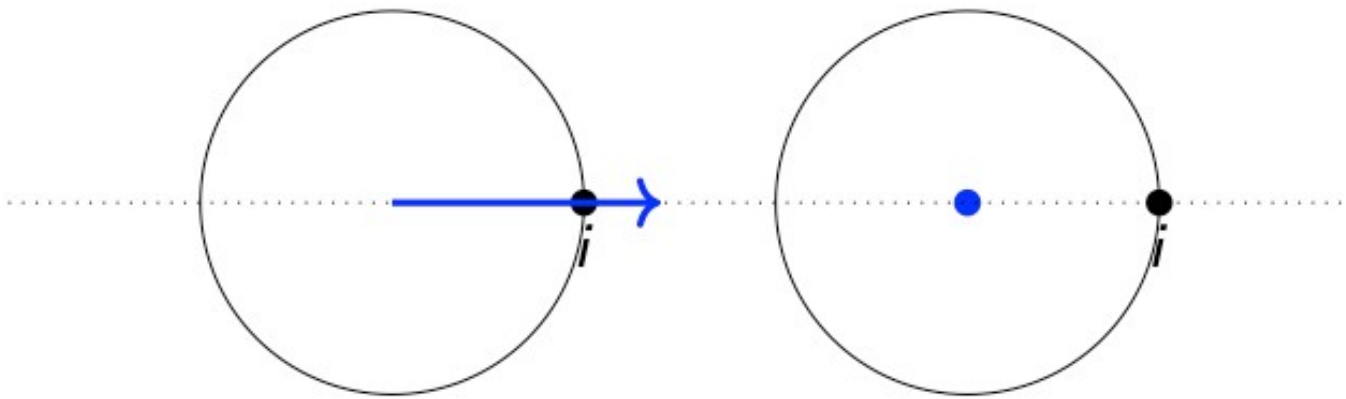


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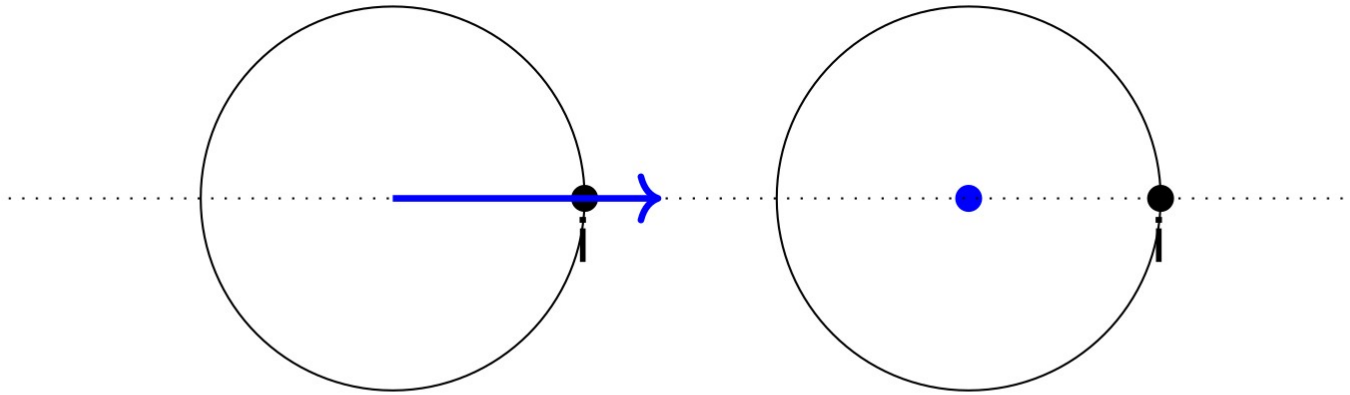


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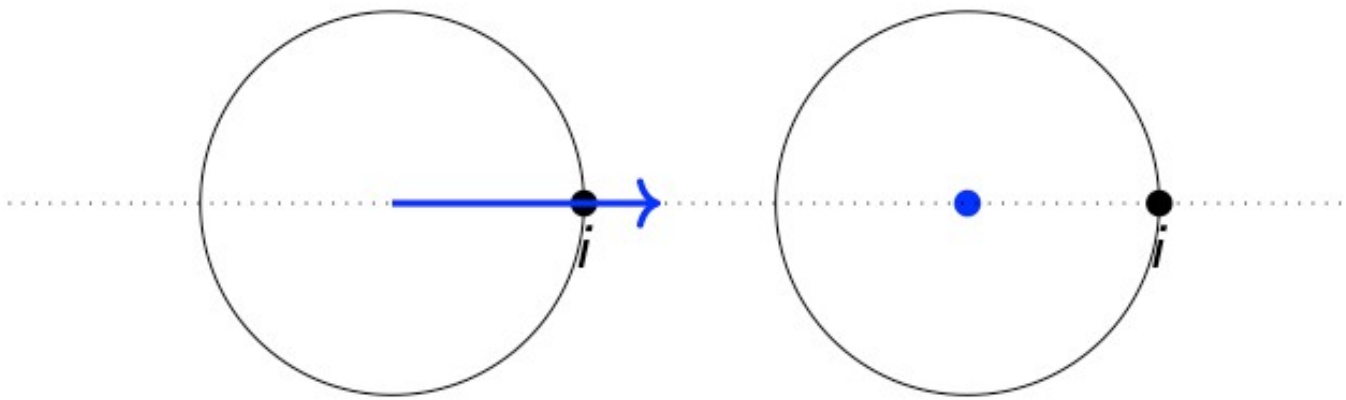
stable

neutrally stable

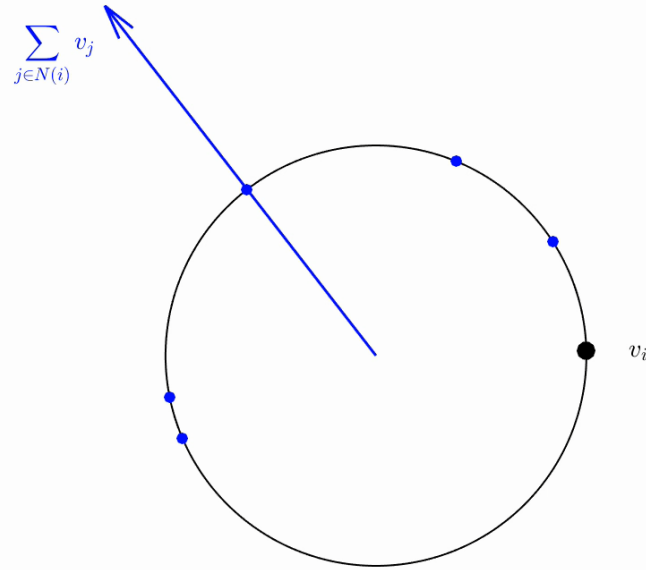
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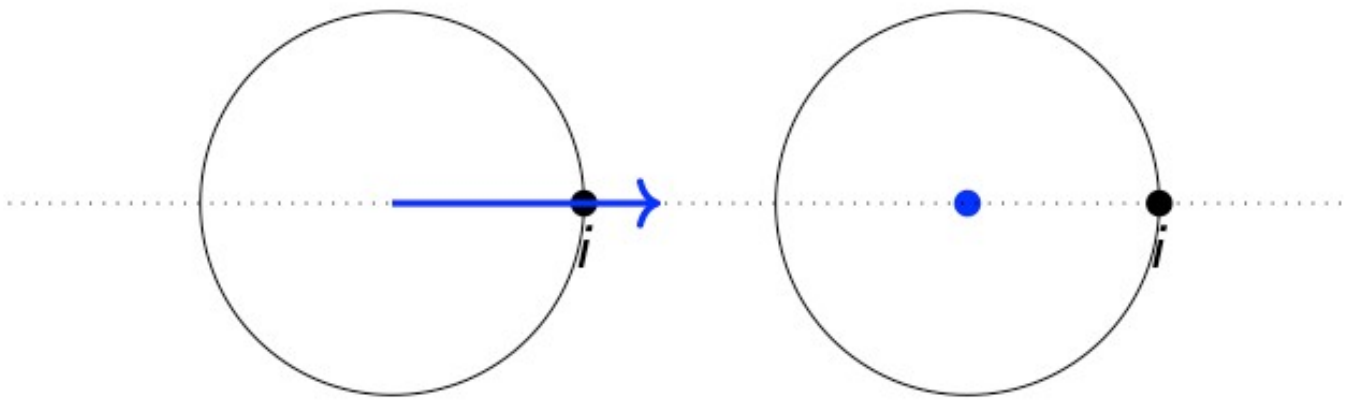


[1] Monzon, Paganini: Global considerations on the Kuramoto model of sinusoidally coupled oscillators

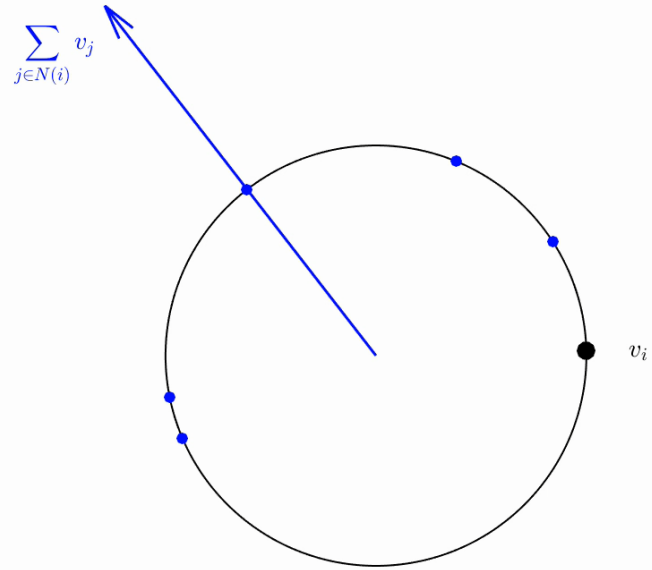
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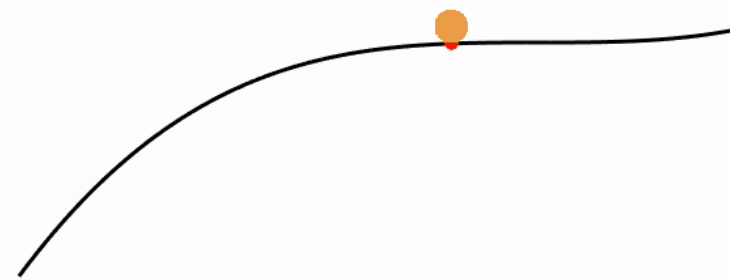
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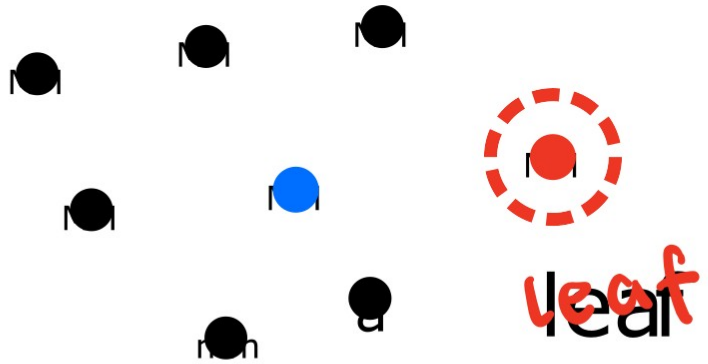




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Our central tool: phasor geometry

Leaf node:



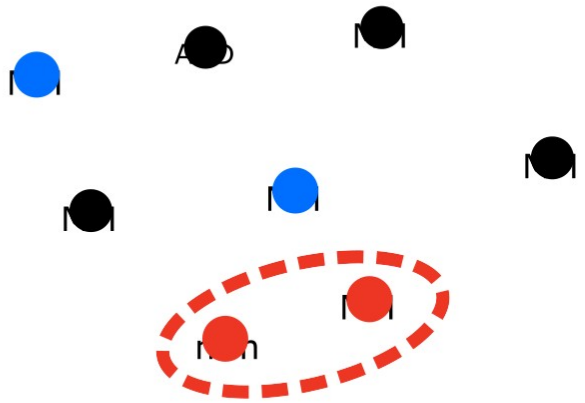
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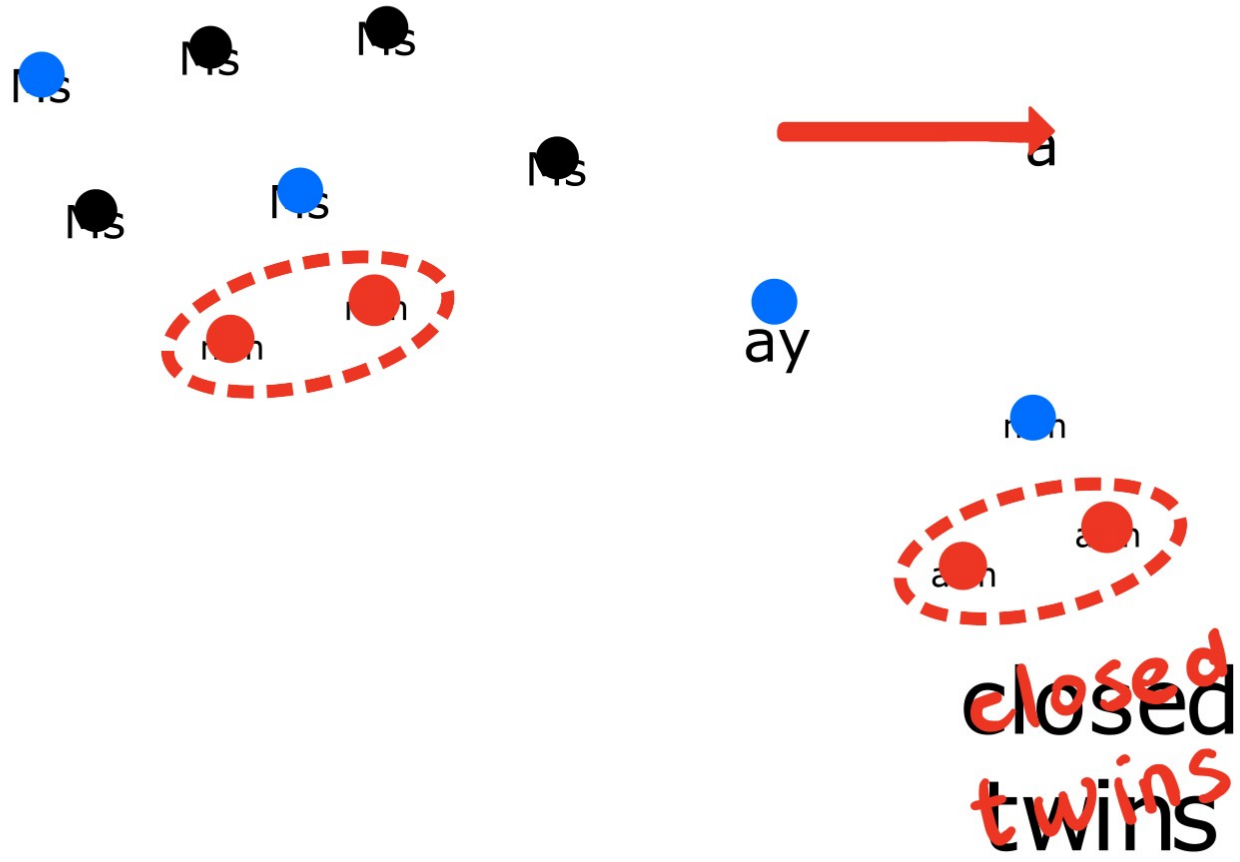
Our central tool: phasor geometry

Structural twins:



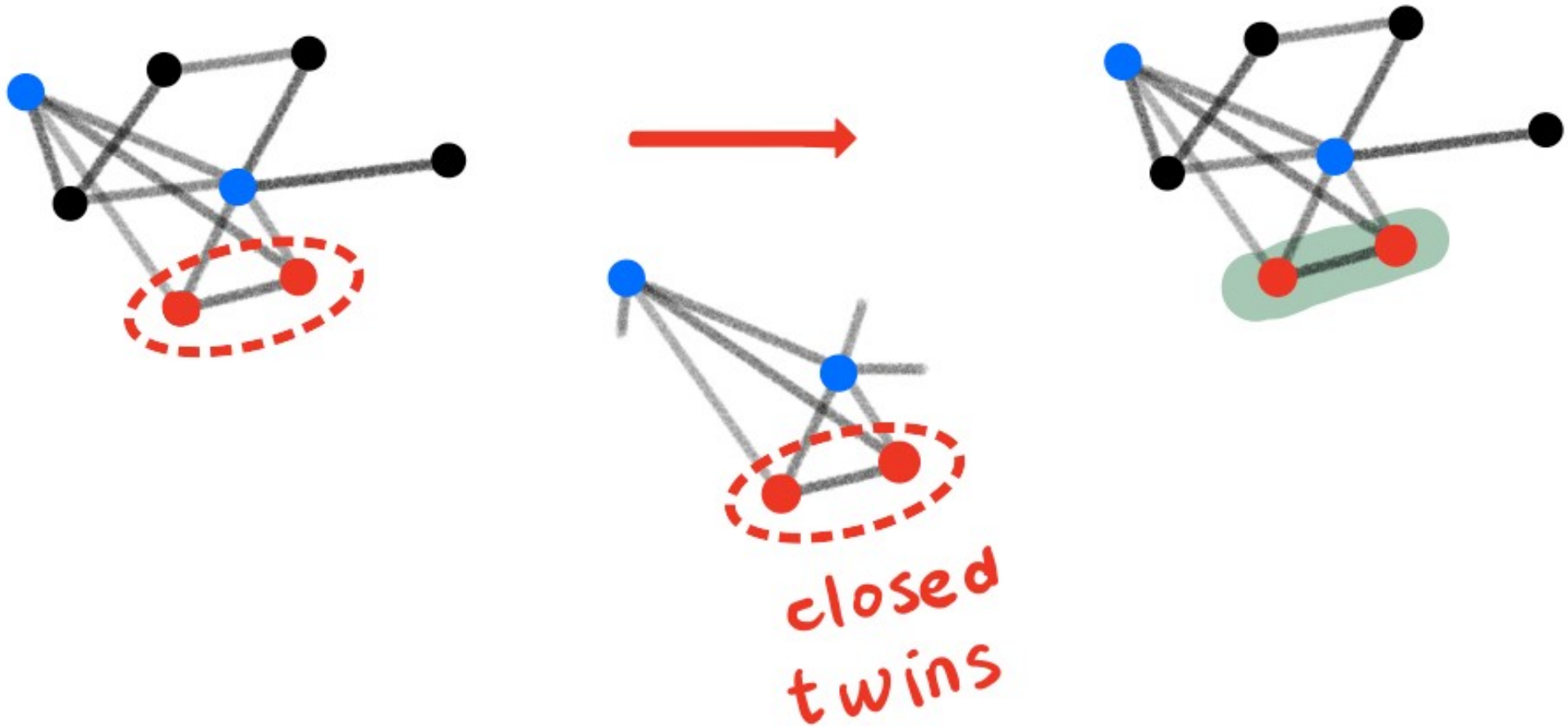
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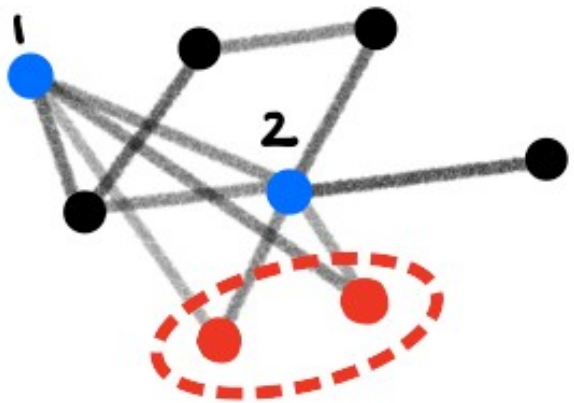
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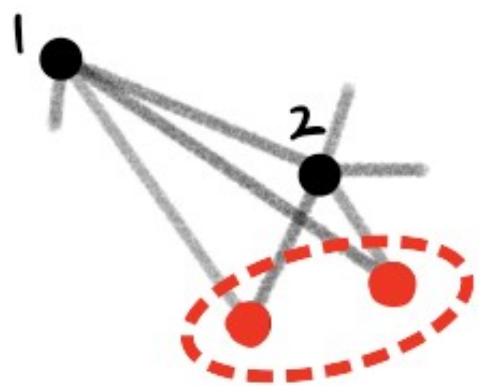
Structural twins:



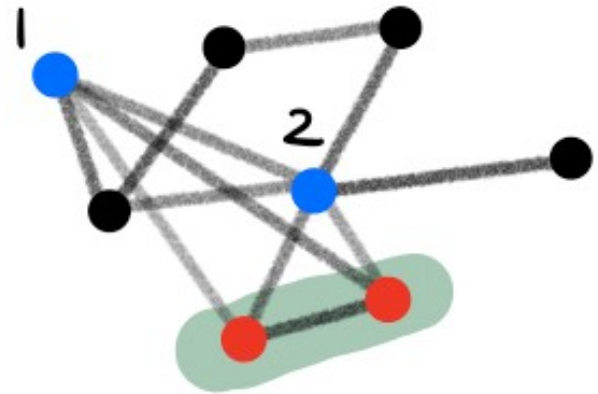
Our central tool: phasor geometry

Structural twins:





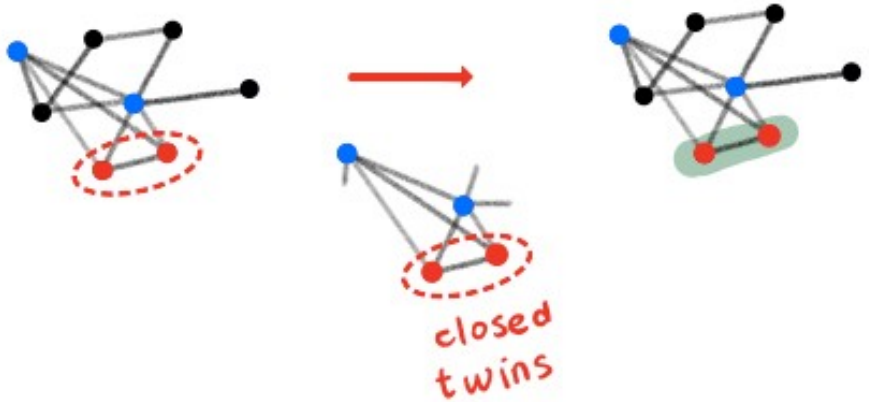
open twins

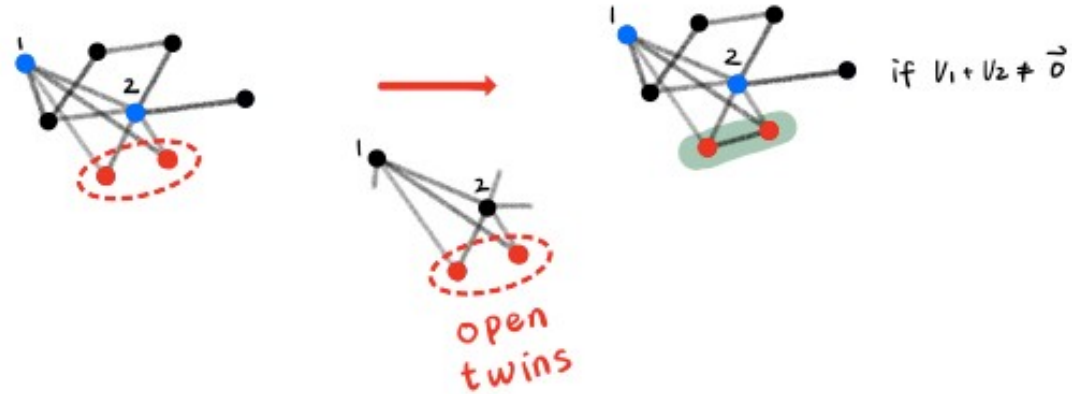


if $V_1 + V_2 \neq \vec{0}$

Our central tool: phasor geometry

Structural twins:





Geometric twins:
A, B are not twins on graph, but still sync,
if the difference between $N(A)$ and $N(V)$ is *nice*.

Result 1: threshold graph $\rightarrow$ quasi-threshold $\rightarrow$ cograph

Theorem

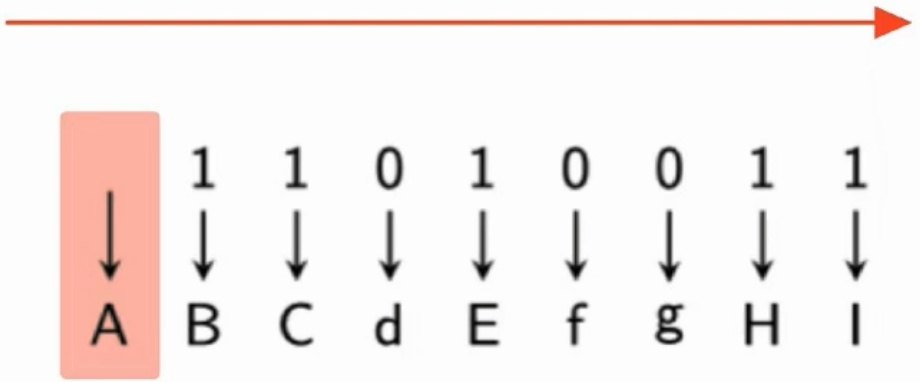
Any connected threshold graph globally synchronizes.

Result 1: threshold graph $\rightarrow$ quasi-threshold $\rightarrow$ cograph

Theorem

Any connected $\boxed{\underbrace{\textit{threshold graph}}_{?}}$ *globally synchronizes.*



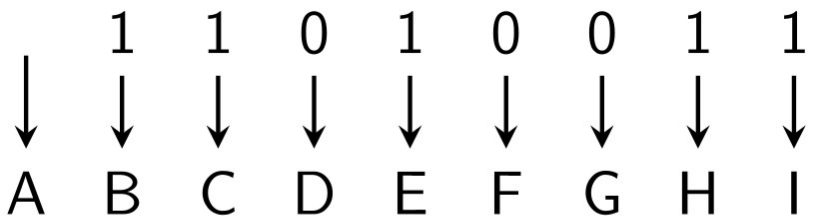


Result 1: threshold graph $\rightarrow$ quasi-threshold $\rightarrow$ cograph

Theorem

Any connected threshold graph globally synchronizes.

- Proof idea.

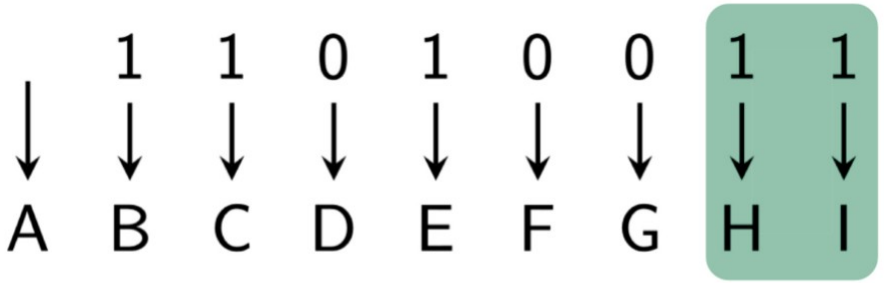


Result 1: threshold graph $\rightarrow$ quasi-threshold $\rightarrow$ cograph

Theorem

Any connected threshold graph globally synchronizes.

- Proof idea. Assume θ is 2nd-order point. closed twins!

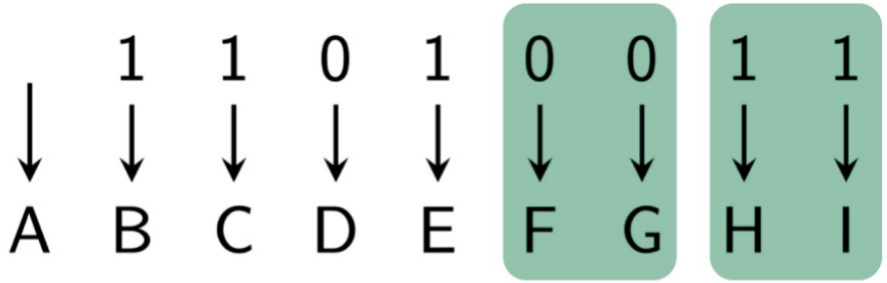


Result 1: threshold graph $\rightarrow$ quasi-threshold $\rightarrow$ cograph

Theorem

Any connected threshold graph globally synchronizes.

- Proof idea. Assume $\boldsymbol{\theta}$ is 2nd-order point. open twins!

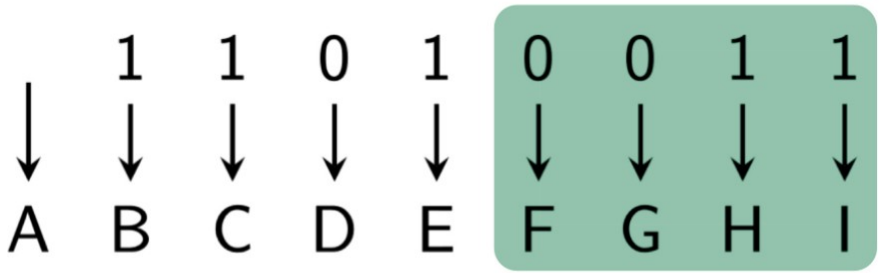


Result 1: threshold graph $\rightarrow$ quasi-threshold $\rightarrow$ cograph

Theorem

Any connected threshold graph globally synchronizes.

- Proof idea. Assume θ is 2nd-order point. merge

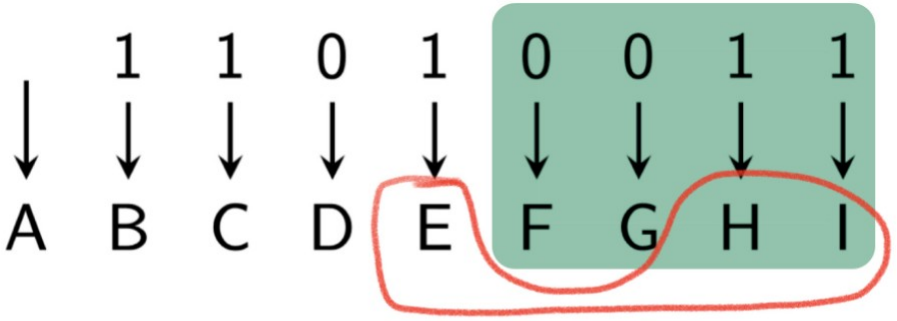


Result 1: threshold graph $\rightarrow$ quasi-threshold $\rightarrow$ cograph

Theorem

Any connected threshold graph globally synchronizes.

- Proof idea. Assume θ is 2nd-order point. geometric twins!

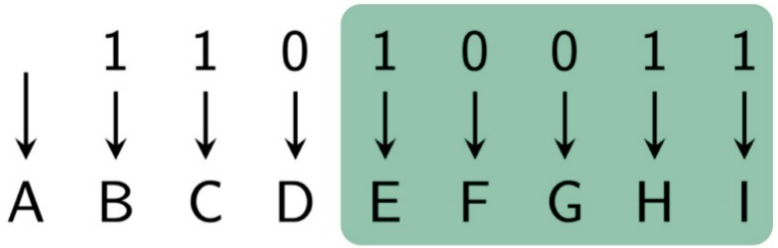


Result 1: threshold graph $\rightarrow$ quasi-threshold $\rightarrow$ cograph

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- Proof idea. Assume $\boldsymbol{\theta}$ is 2nd-order point.

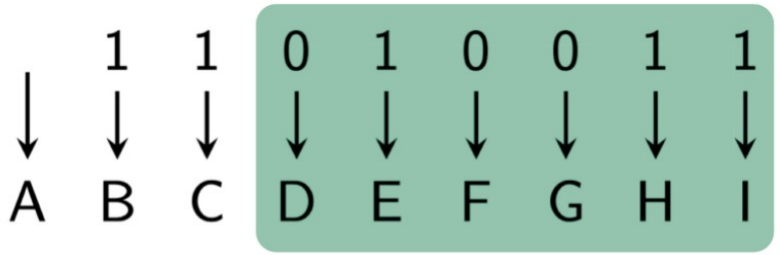


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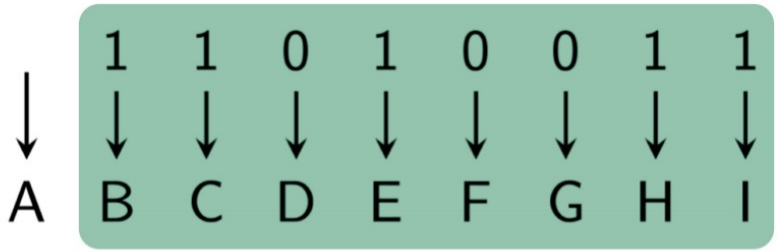


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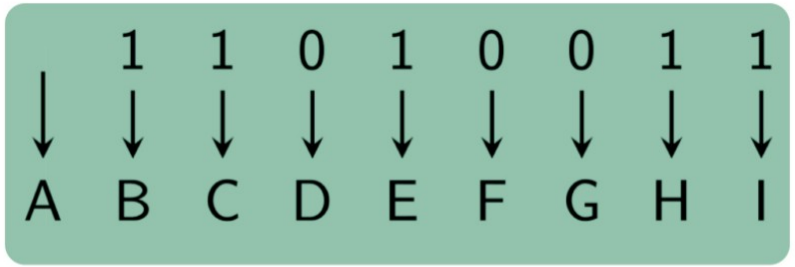


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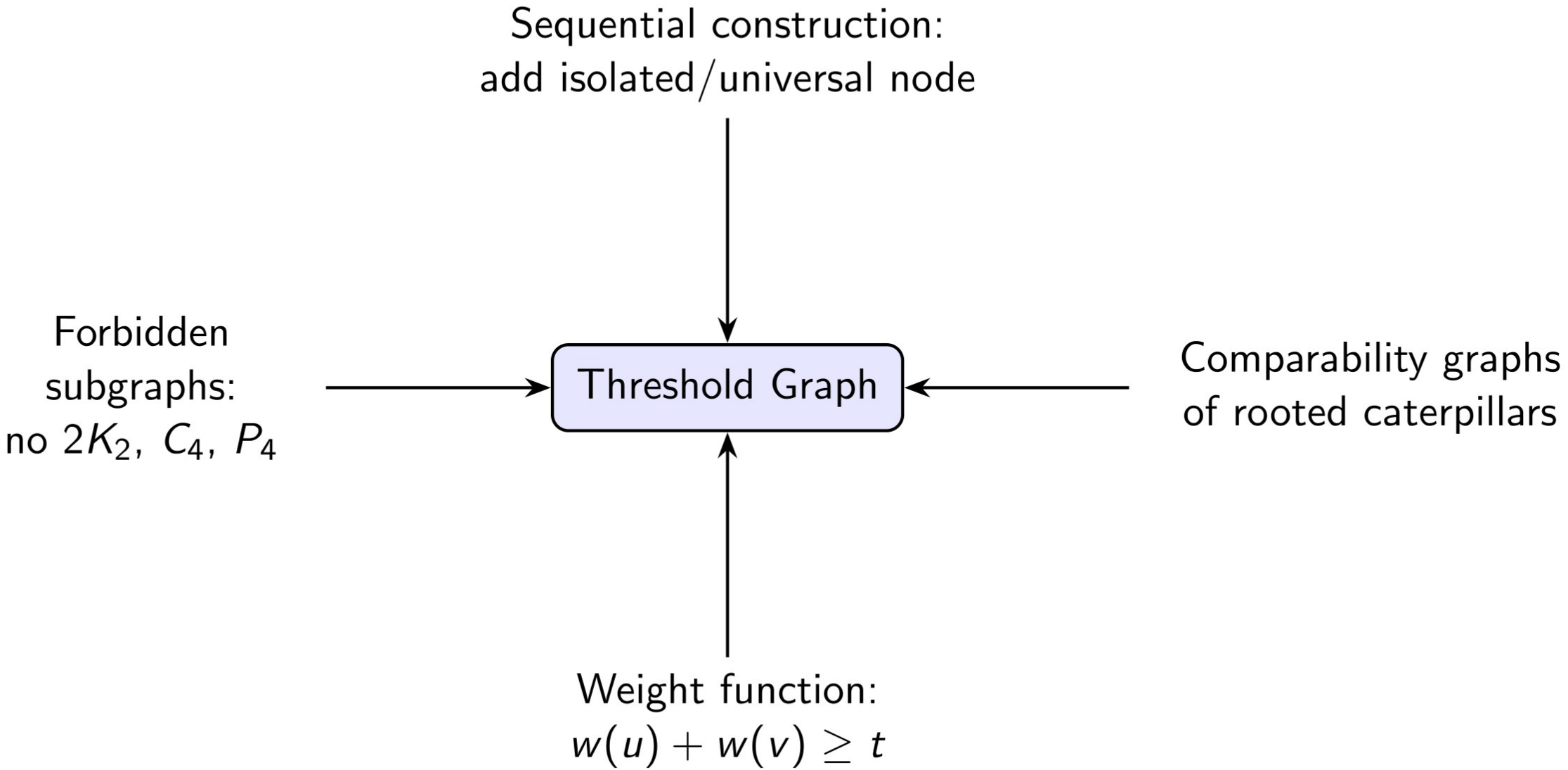
Theorem

Any connected threshold graph globally synchronizes.

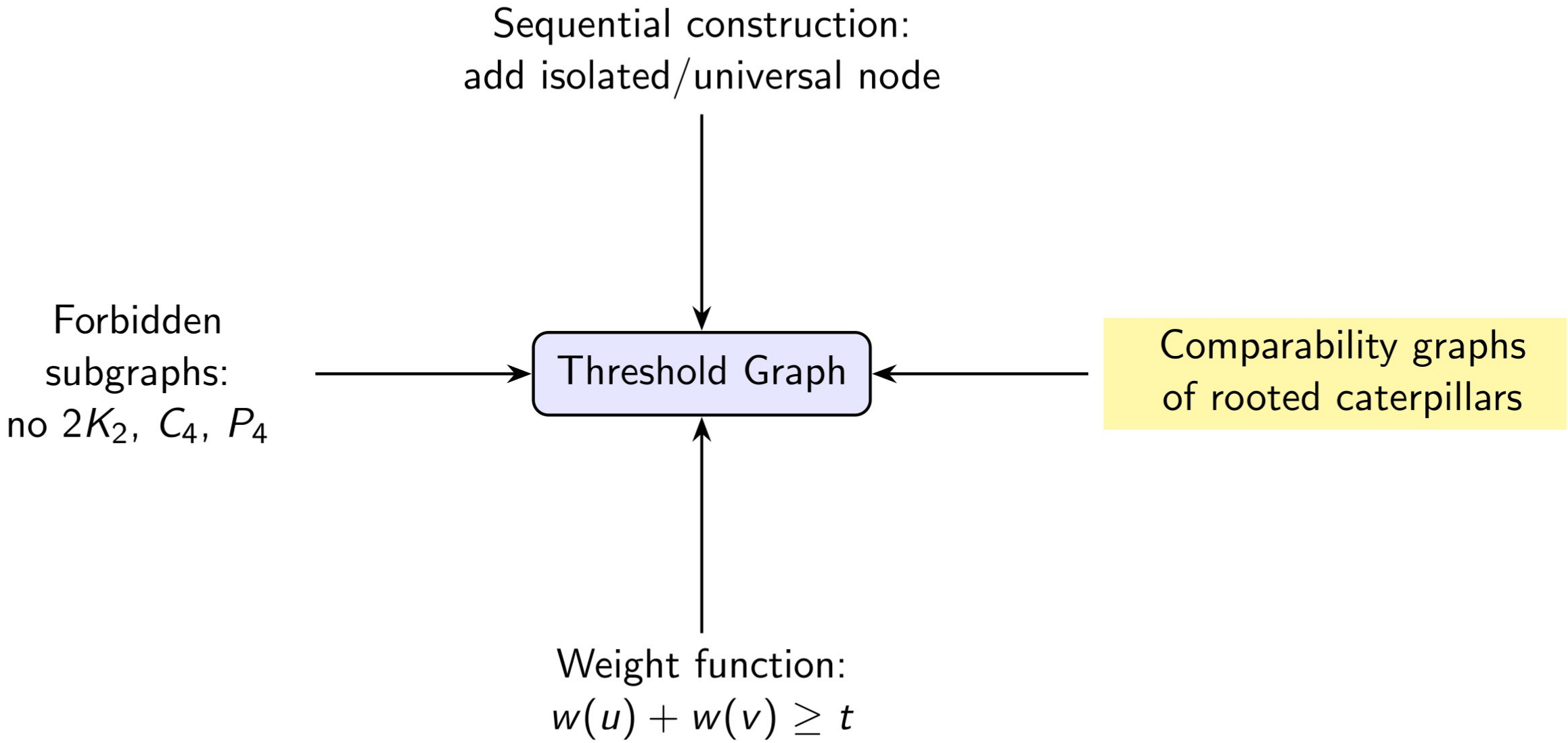
- Proof idea. Assume $\boldsymbol{\theta}$ is 2nd-order point. all nodes sync



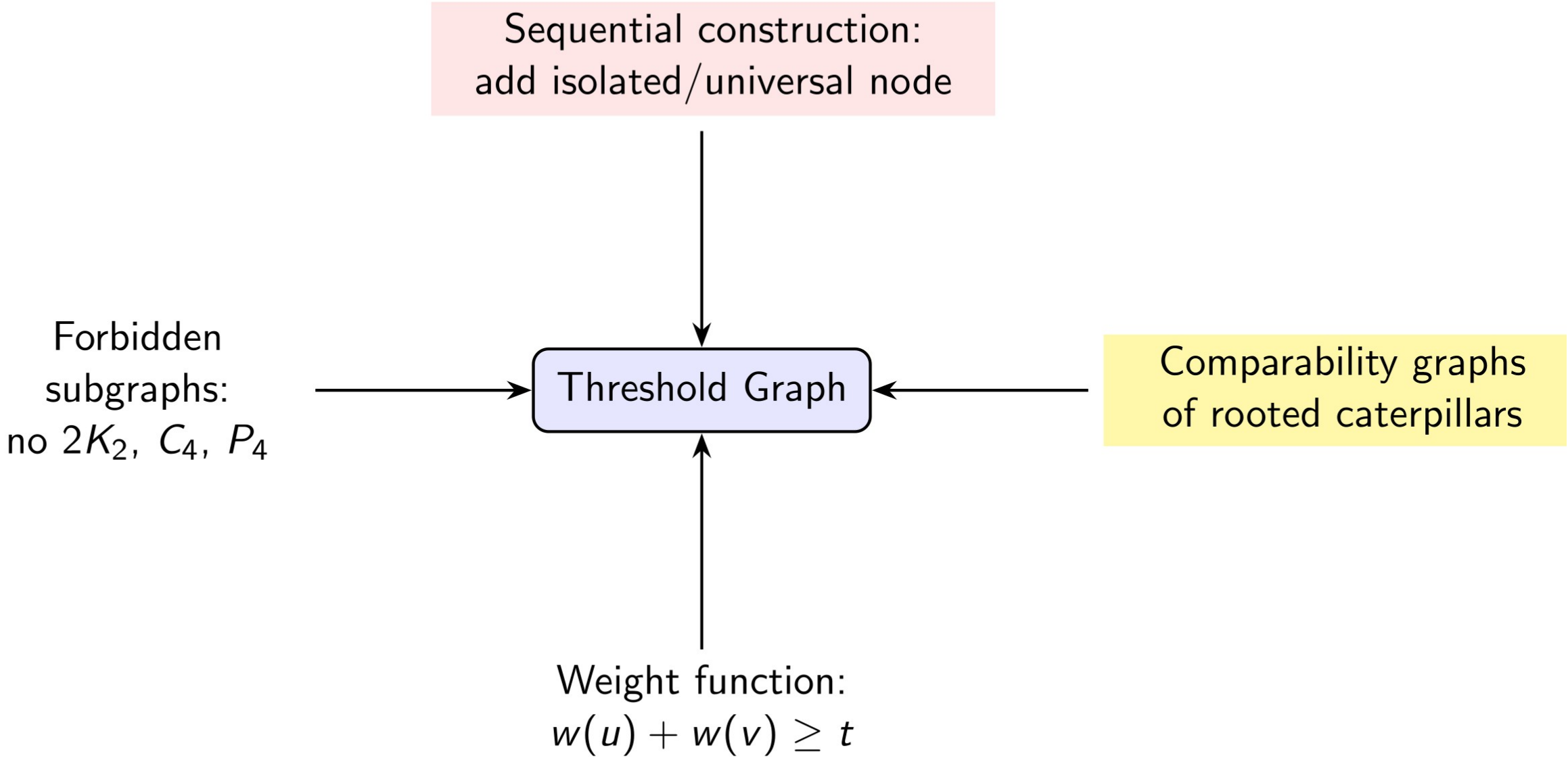
Simple but rich structure!



Simple but rich structure!



Simple but rich structure!



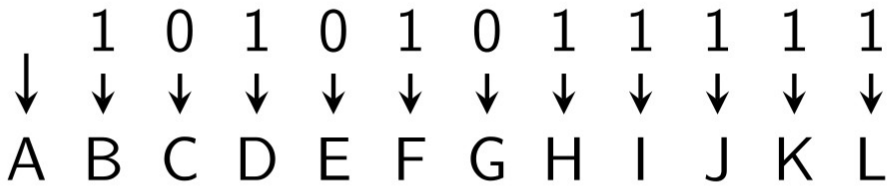
Result 1: threshold graph → quasi-threshold → cograph

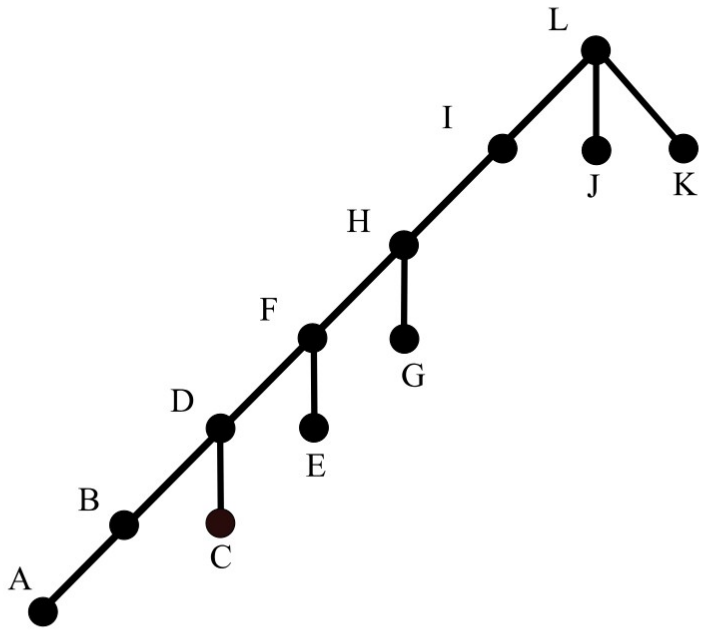
Threshold graphs have another chacterization (Def 2), and they global sync!

Def 1: Closed under

1. isolated node
2. universal node

Def 2: Comparability graphs of rooted caterpillars





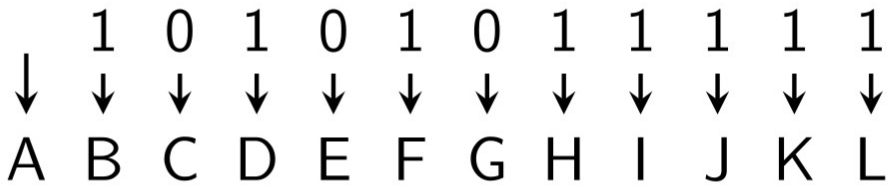
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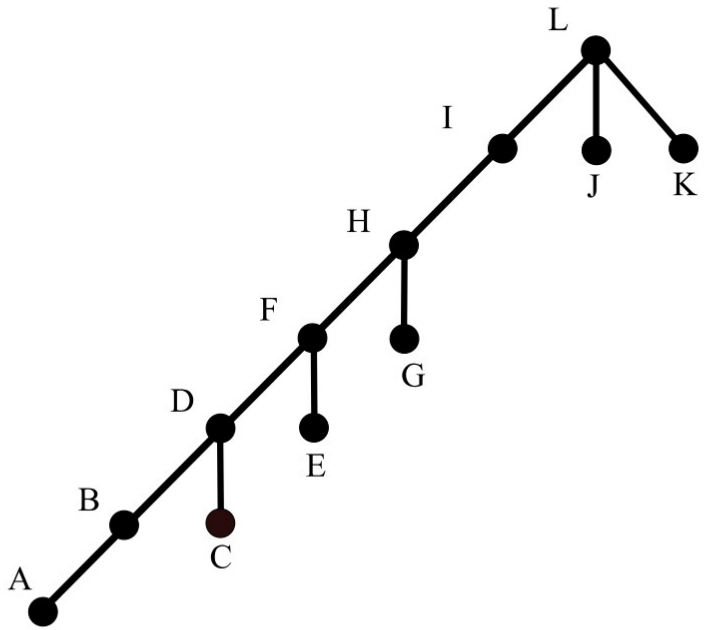
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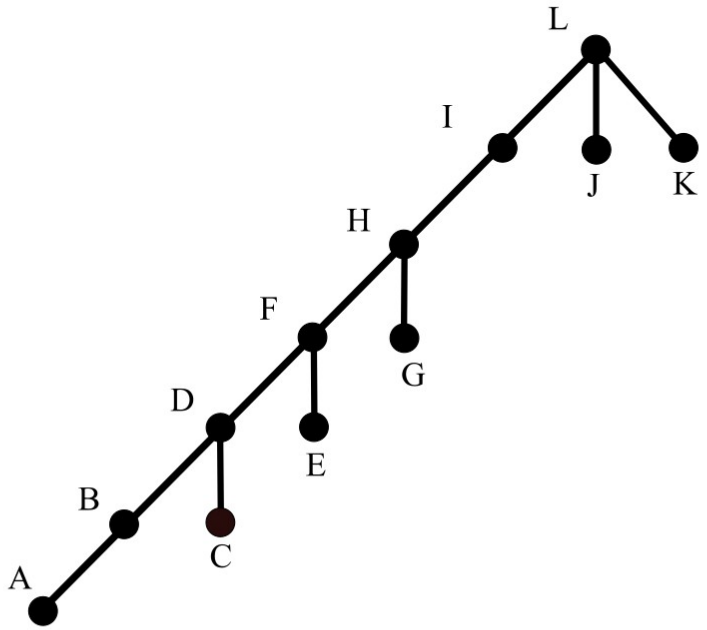
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Def 2: Comparability graphs of rooted caterpillars

?

	1	0	1	0	1	0	1	1	1	1	1
↓	↓	↓	↓	↓	↓	↓	↓	↓	↓	↓	↓
A	B	C	D	E	F	G	H	I	J	K	L



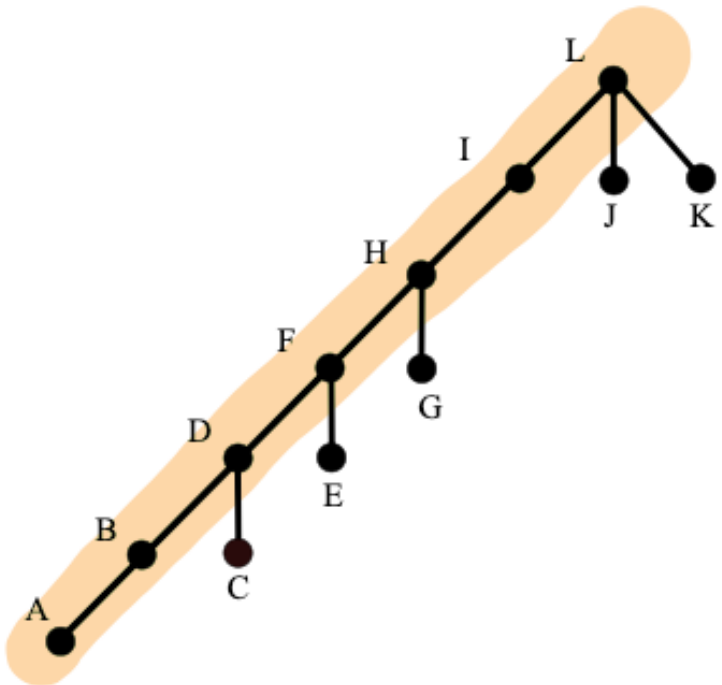
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Def 2: Comparability graphs of rooted caterpillars



	1	0	1	0	1	0	1	1	1	1	1
↓	↓	↓	↓	↓	↓	↓	↓	↓	↓	↓	↓
A	B	C	D	E	F	G	H	I	J	K	L

Result 1: threshold graph → quasi-threshold → cograph

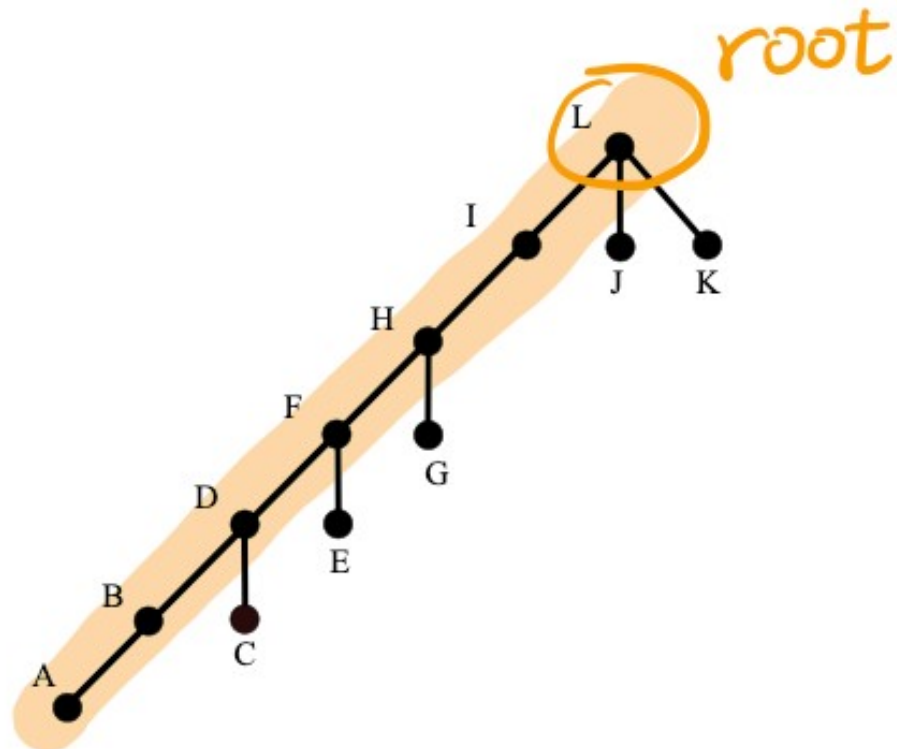
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↓	↓	↓	↓	↓	↓	↓	↓	↓	↓	↓	↓
A	B	C	D	E	F	G	H	I	J	K	L

Def 2: Comparability graphs of rooted caterpillars



Result 1: threshold graph → quasi-threshold → cograph

Threshold graphs have another chacterization (Def 2), and they global sync!

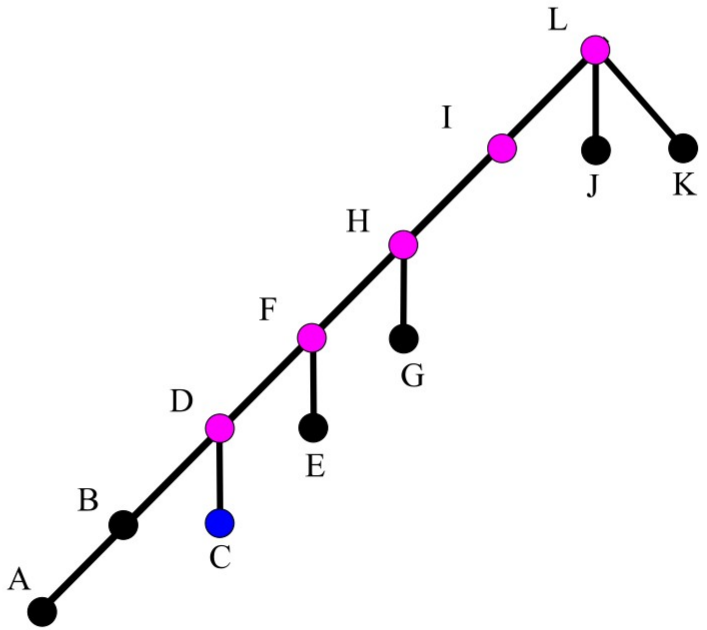
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Def 2: Comparability graphs of rooted caterpillars

?

	1	0	1	0	1	0	1	1	1	1	1
↓	↓	↓	↓	↓	↓	↓	↓	↓	↓	↓	↓
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Result 1: threshold graph → quasi-threshold → cograph

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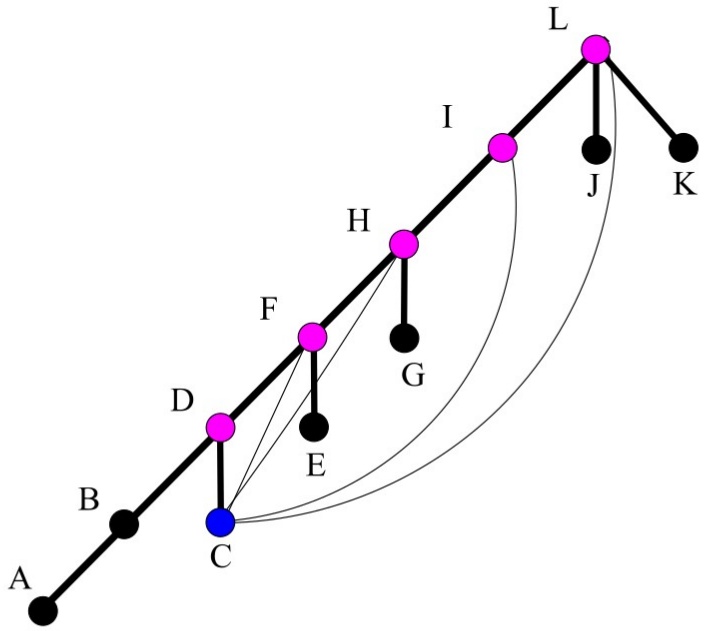
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↓	↓	↓	↓	↓	↓	↓	↓	↓	↓	↓	↓
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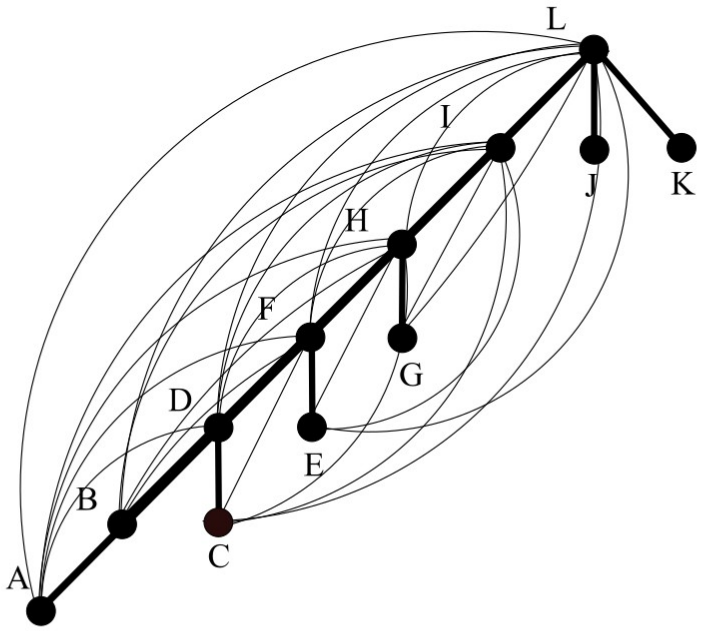
Def 1: Closed under

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Def 2: Comparability graphs of rooted caterpillars

?

	1	0	1	0	1	0	1	1	1	1	1
↓	↓	↓	↓	↓	↓	↓	↓	↓	↓	↓	↓
A	B	C	D	E	F	G	H	I	J	K	L



Result 1: threshold graph → quasi-threshold → cograph

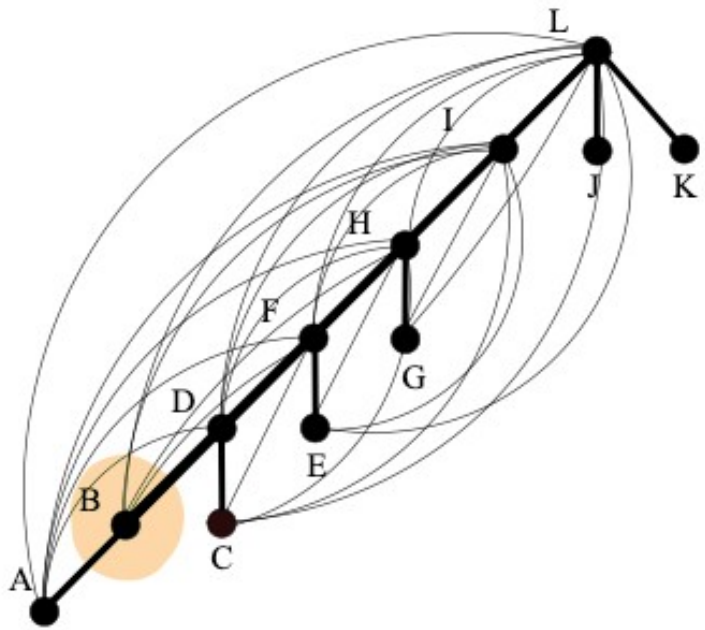
Threshold graphs have another chacterization (Def 2), and they global sync!

Def 1: Closed under

1. add isolated node
2. add universal node

Def 2: Comparability graphs of rooted caterpillars

	1	0	1	0	1	0	1	1	1	1	1
↓	↓	↓	↓	↓	↓	↓	↓	↓	↓	↓	↓
A	B	C	D	E	F	G	H	I	J	K	L



Result 1: threshold graph → quasi-threshold → cograph

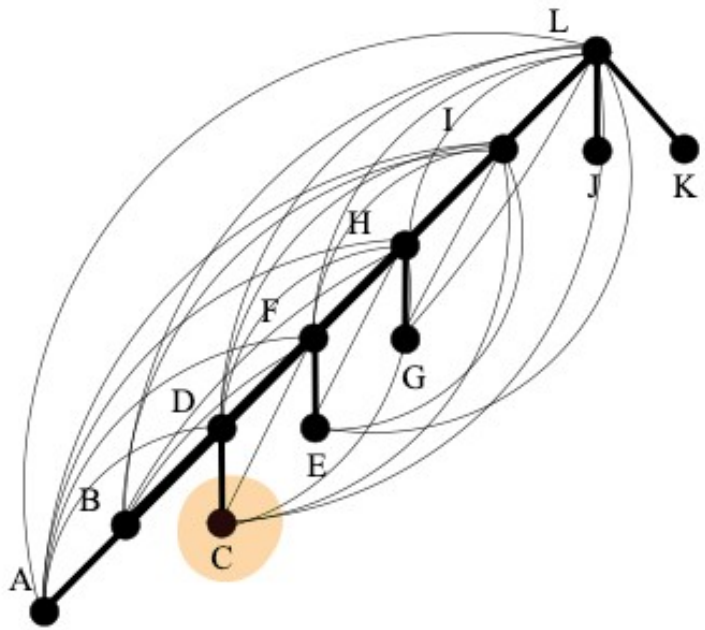
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	1	0	1	0	1	0	1	1	1	1	1
↓	↓	↓	↓	↓	↓	↓	↓	↓	↓	↓	↓
A	B	C	D	E	F	G	H	I	J	K	L



Result 1: threshold graph $\rightarrow$ quasi-threshold $\rightarrow$ cograph

Does the comparability graph of a rooted tree also synchronize globally?

Result 1: threshold graph $\rightarrow$ quasi-threshold $\rightarrow$ cograph

Does the comparability graph of a rooted tree also synchronize globally? ✔

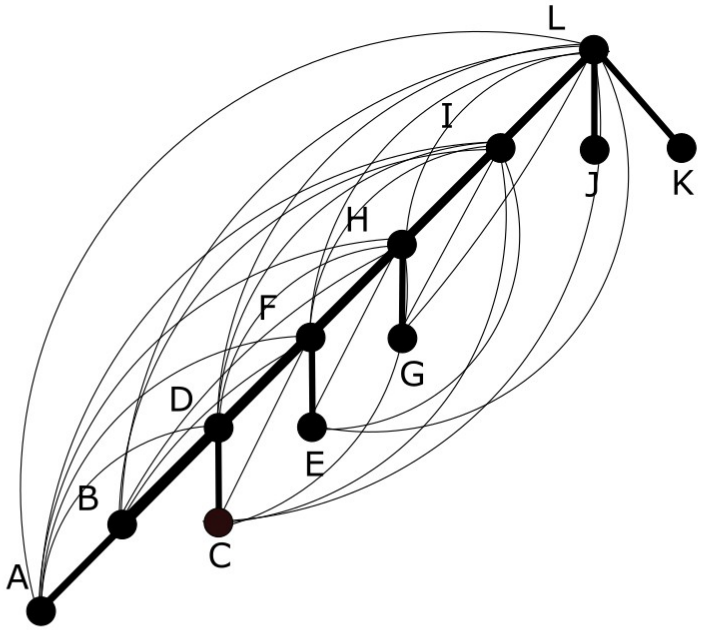
Result 1: threshold graph → quasi-threshold → cograph

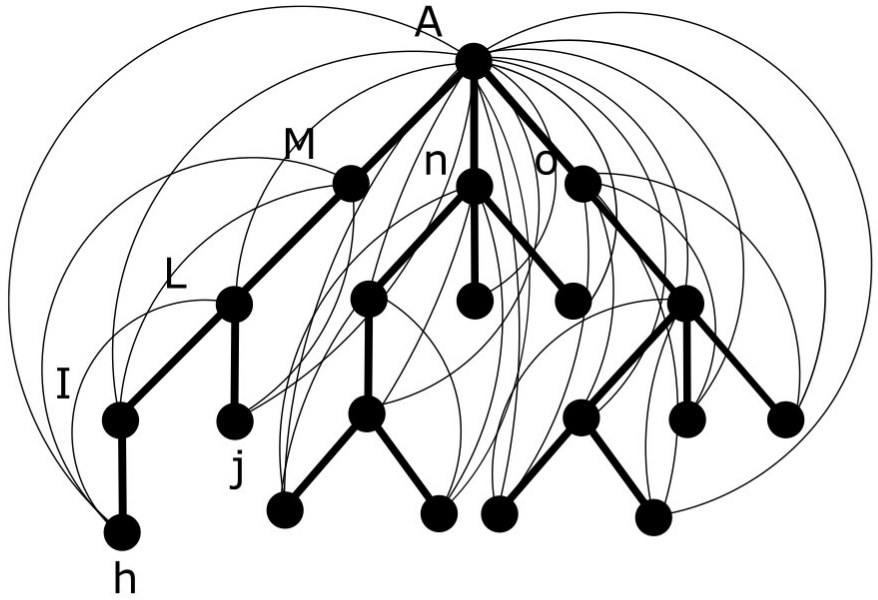
Does the comparability graph of a rooted tree also synchronize globally? ✓

Comp. graph of rooted caterpillar (threshold graphs).

Comp. graph of rooted tree (Quasi-threshold graphs).

ropagation





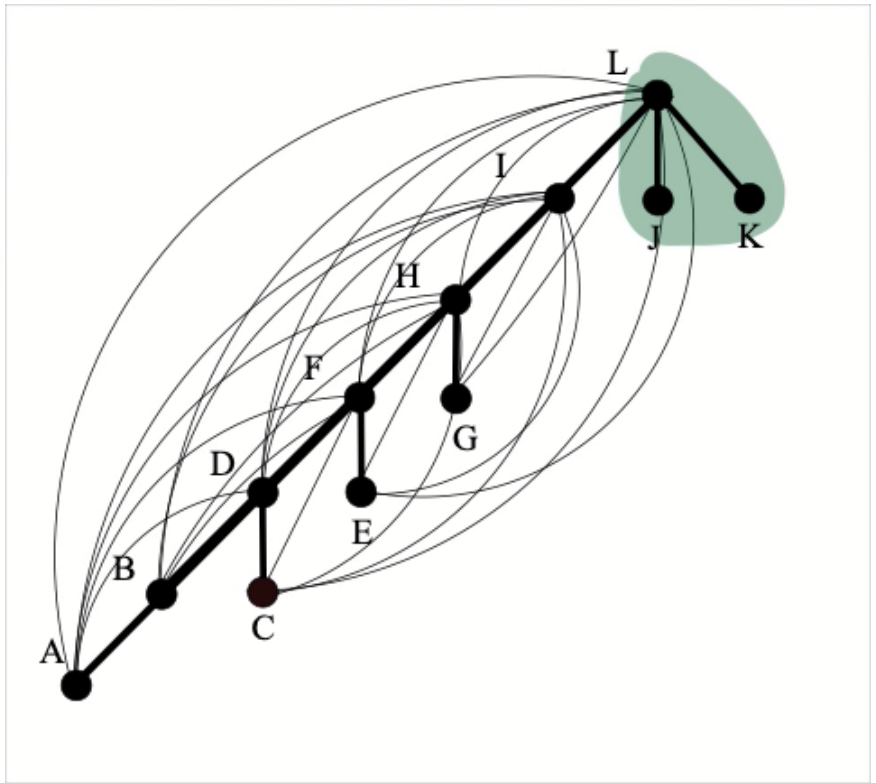
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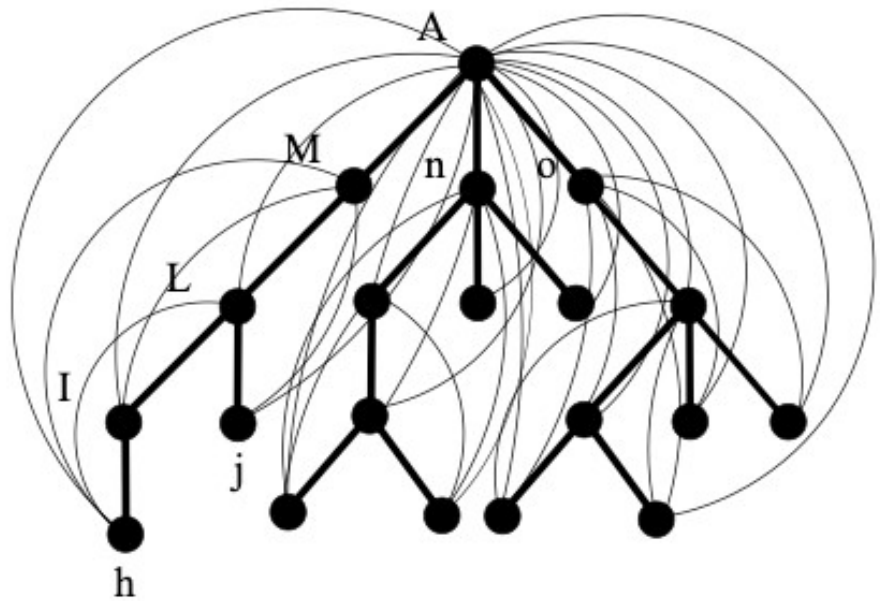
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Comp. graph of rooted caterpillar (threshold graphs).

Comp. graph of rooted tree (Quasi-threshold graphs).

propagation

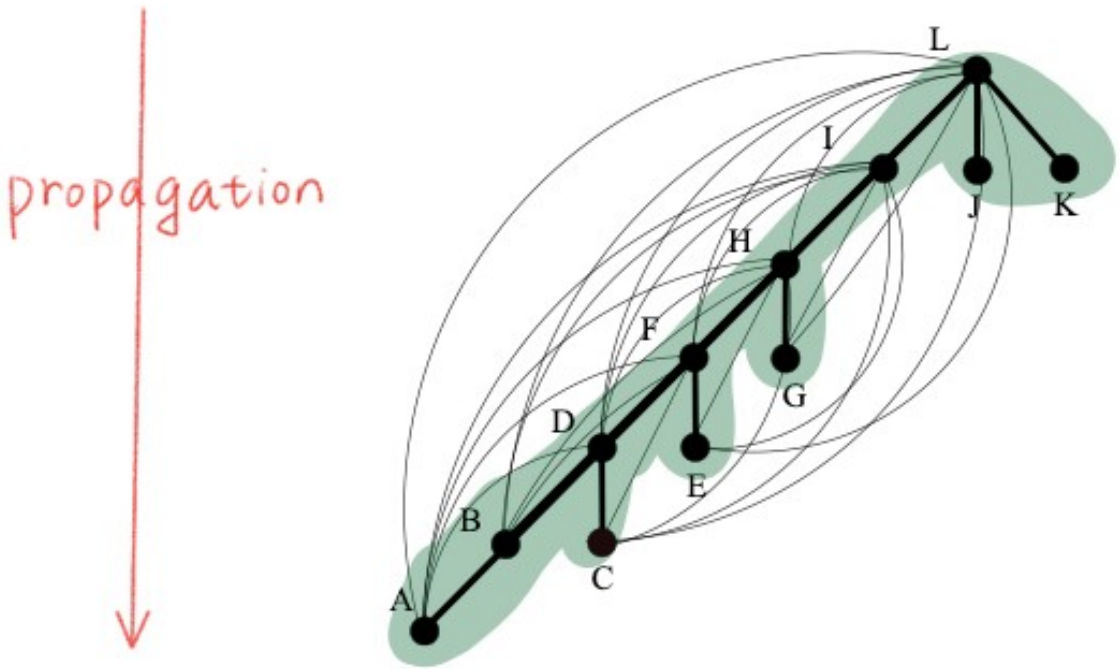




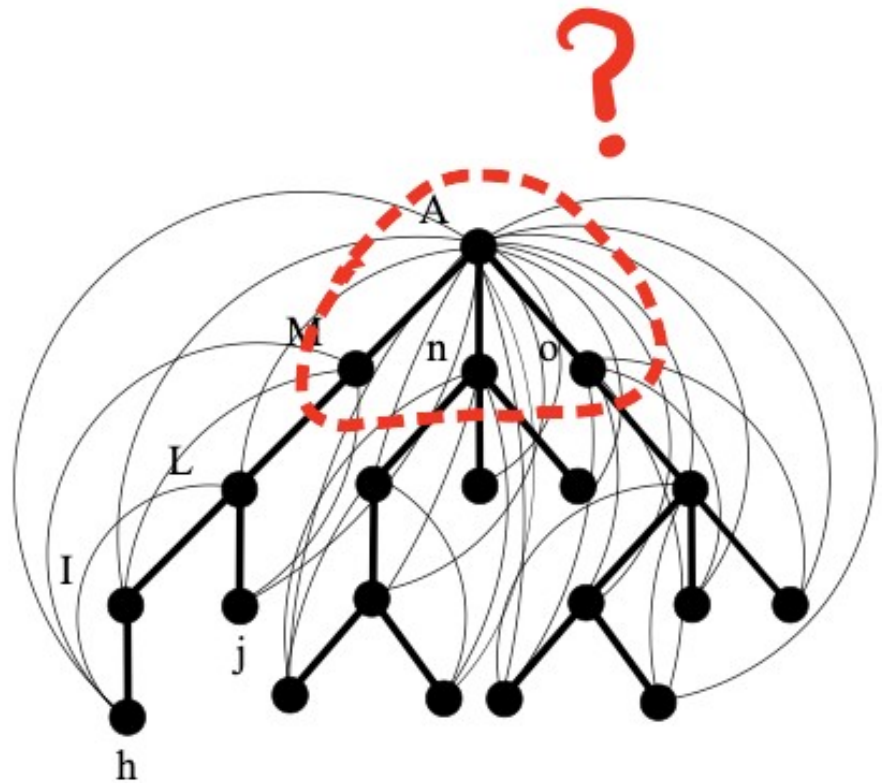
Result 1: threshold graph → quasi-threshold → cograph

Does the comparability graph of a rooted tree also synchronize globally? ✔

Comp. graph of rooted caterpillar (threshold graphs).



Comp. graph of rooted tree (Quasi-threshold graphs).



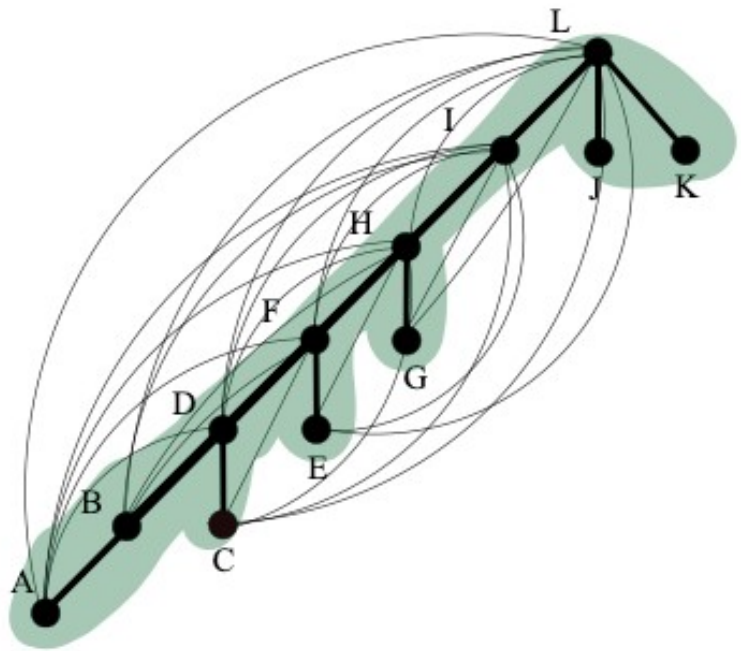
Result 1: threshold graph → quasi-threshold → cograph

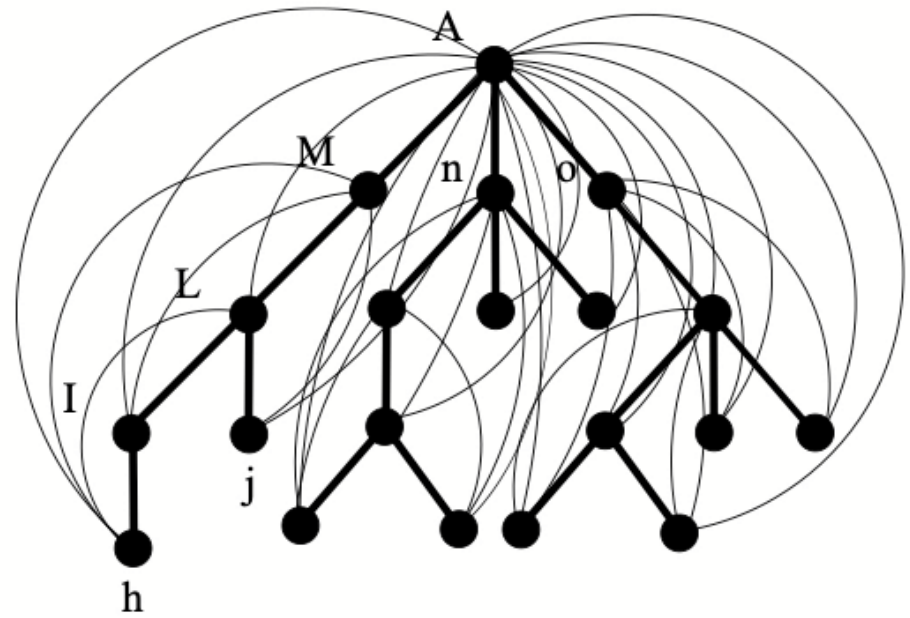
Does the comparability graph of a rooted tree also synchronize globally? ✔

Comp. graph of rooted caterpillar (threshold graphs).

Comp. graph of rooted tree (Quasi-threshold graphs).

propagation



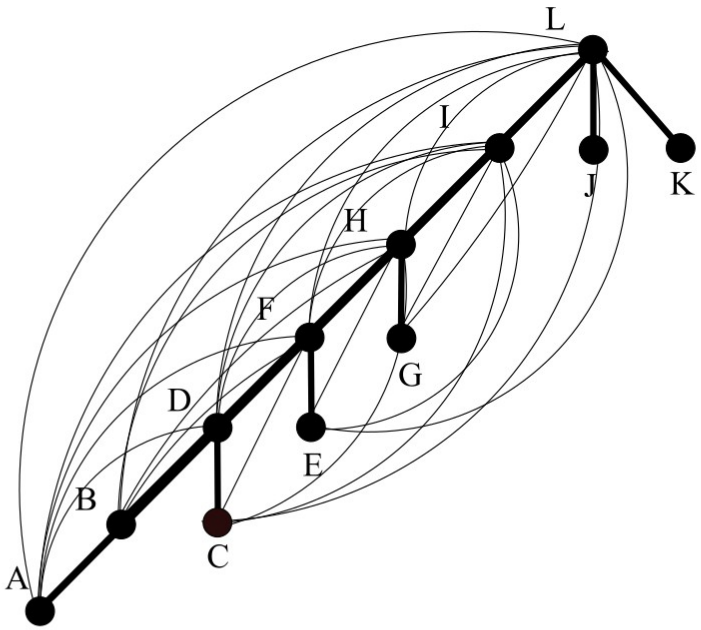


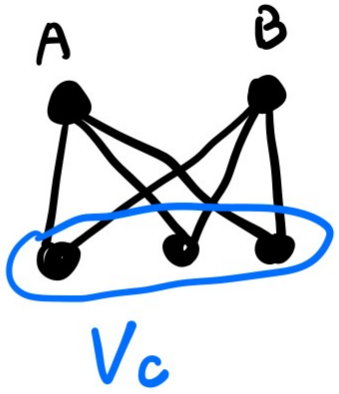
Result 1: threshold graph $\rightarrow$ quasi-threshold $\rightarrow$ cograph

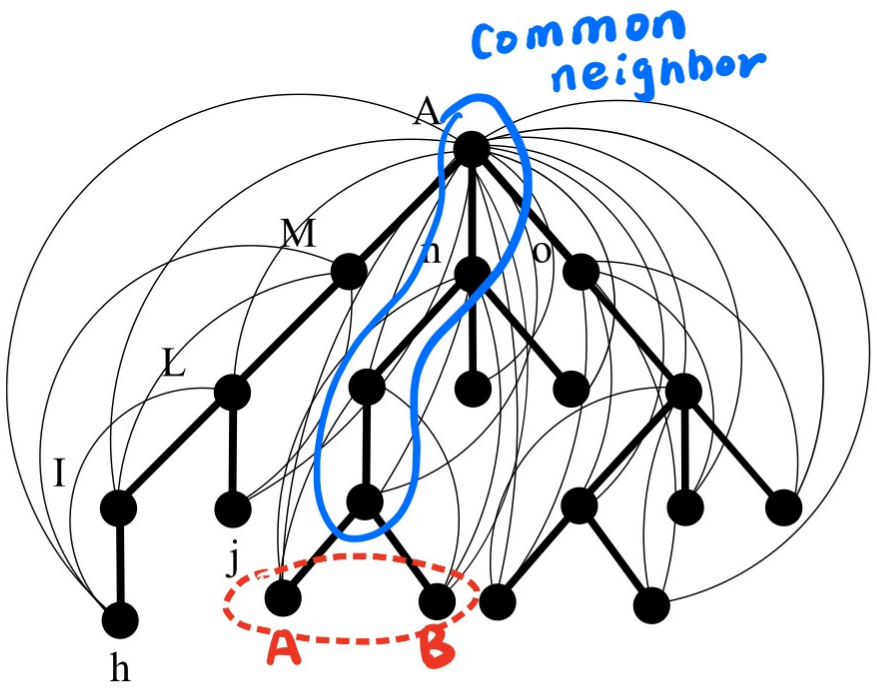
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Comp. graph of rooted caterpillar (threshold graphs).

Comp. graph of rooted tree (Quasi-threshold graphs).



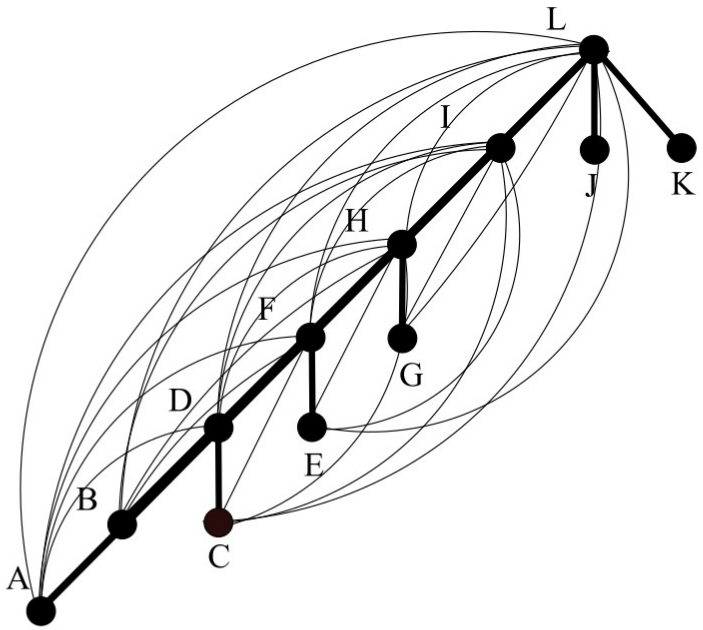




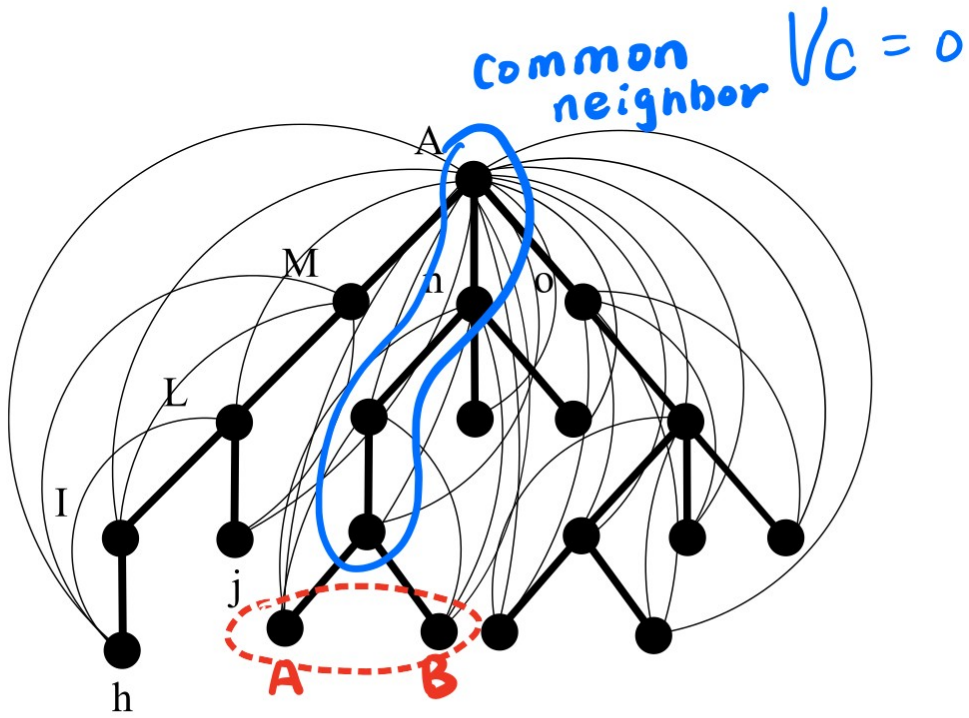
Result 1: threshold graph $\rightarrow$ quasi-threshold $\rightarrow$ cograph

Does the comparability graph of a rooted tree also synchronize globally? ✔

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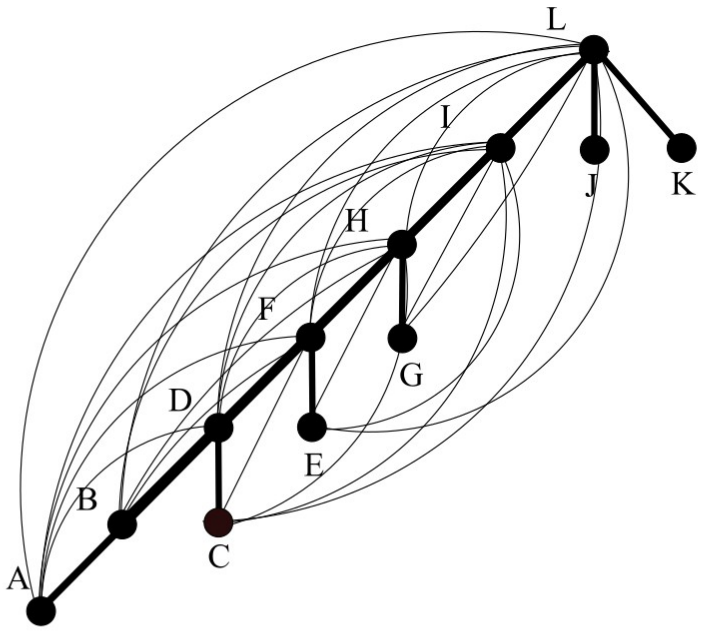


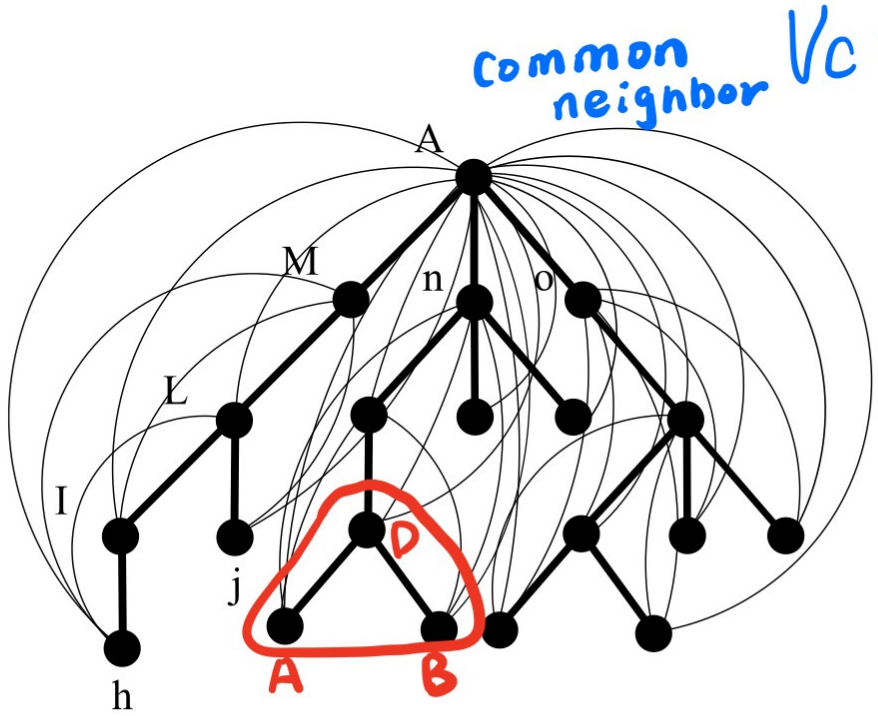
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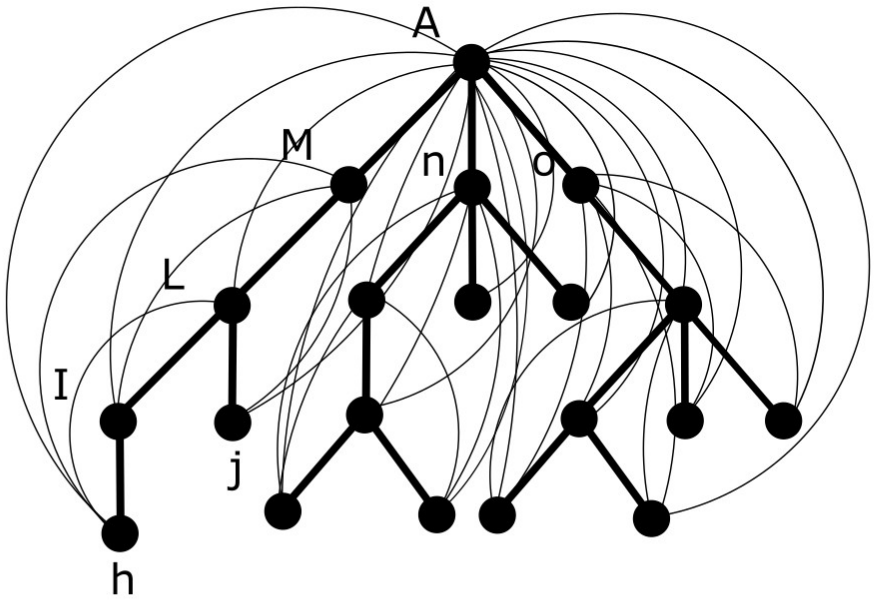


ABD as a group receiving antipodal force

Result 1: threshold graph → quasi-threshold → cograph

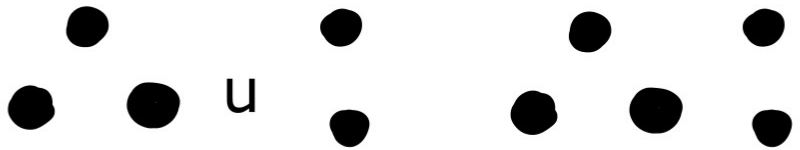
Does the comparability graph of a rooted tree also synchronize globally? ✓

Def 1: Comp. graph of rooted tree



Def 2: Closed under

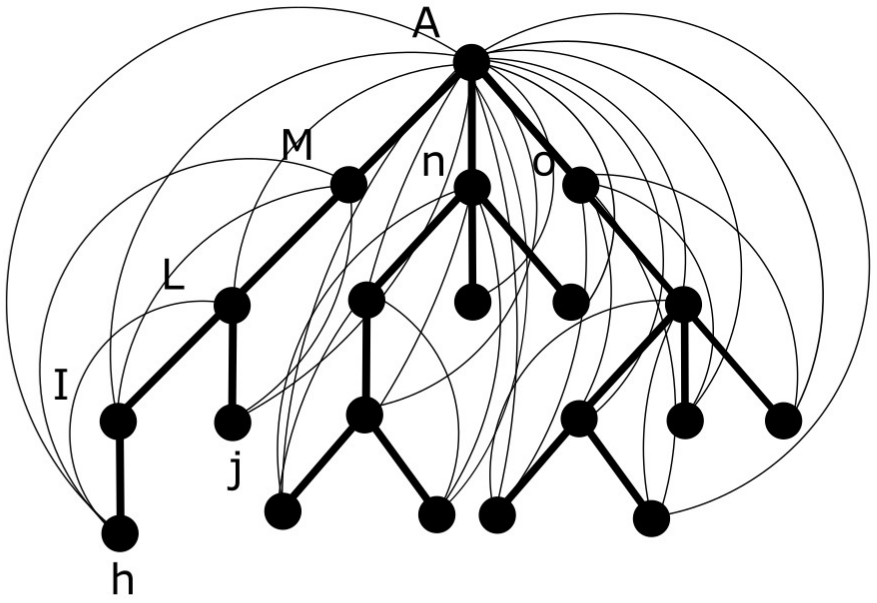
1. disjoint union
2. add universal node



Result 1: threshold graph → quasi-threshold → cograph

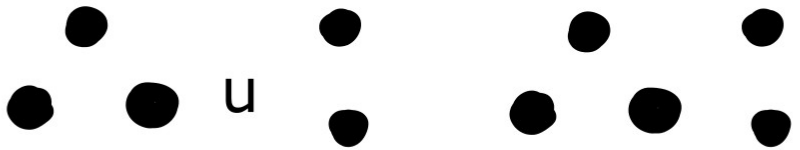
Does the comparability graph of a rooted tree also synchronize globally? ✓

Def 1: Comp. graph of rooted tree



Def 2: Closed under

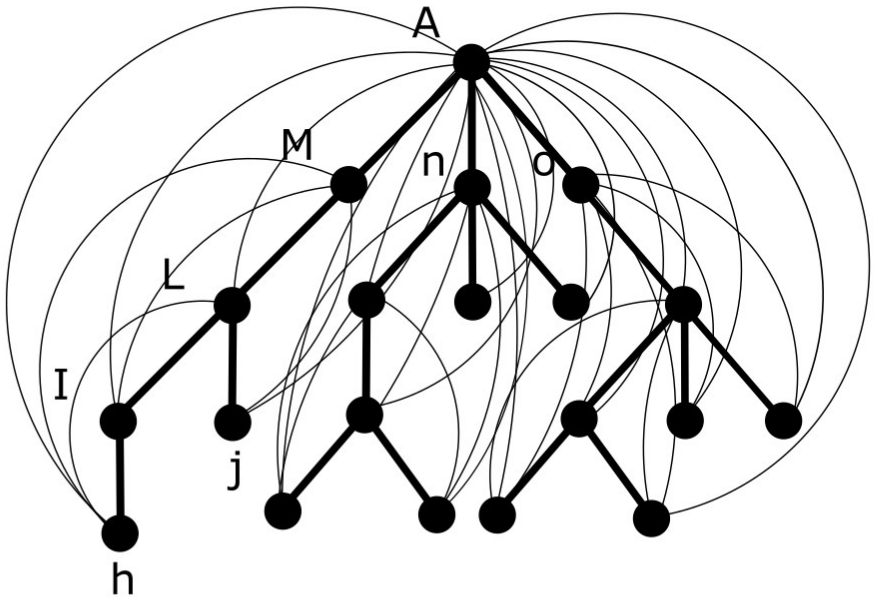
1. disjoint union (← generalize ——— add isolated node)
2. add universal node



Result 1: threshold graph → quasi-threshold → cograph

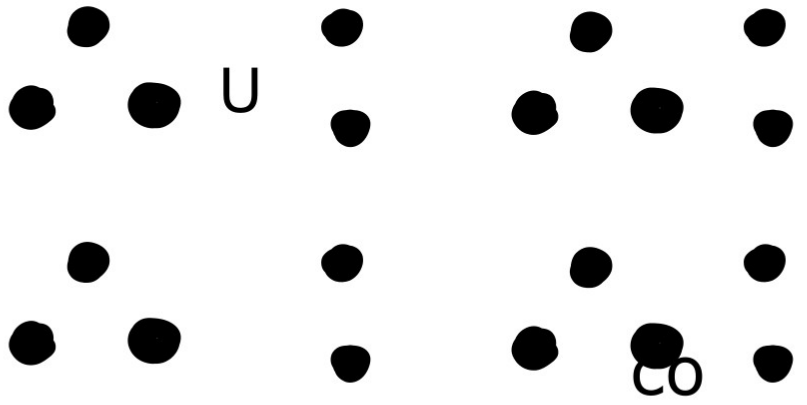
Does the comparability graph of a rooted tree also synchronize globally? ✓

Def 1: Comp. graph of rooted tree



Def 2: Closed under

1. disjoint union (← generalize ----- add isolated node)
2. add universal node ---- generalize ----→ join



? Does graph constructed by {disjoint union, join} global sync?

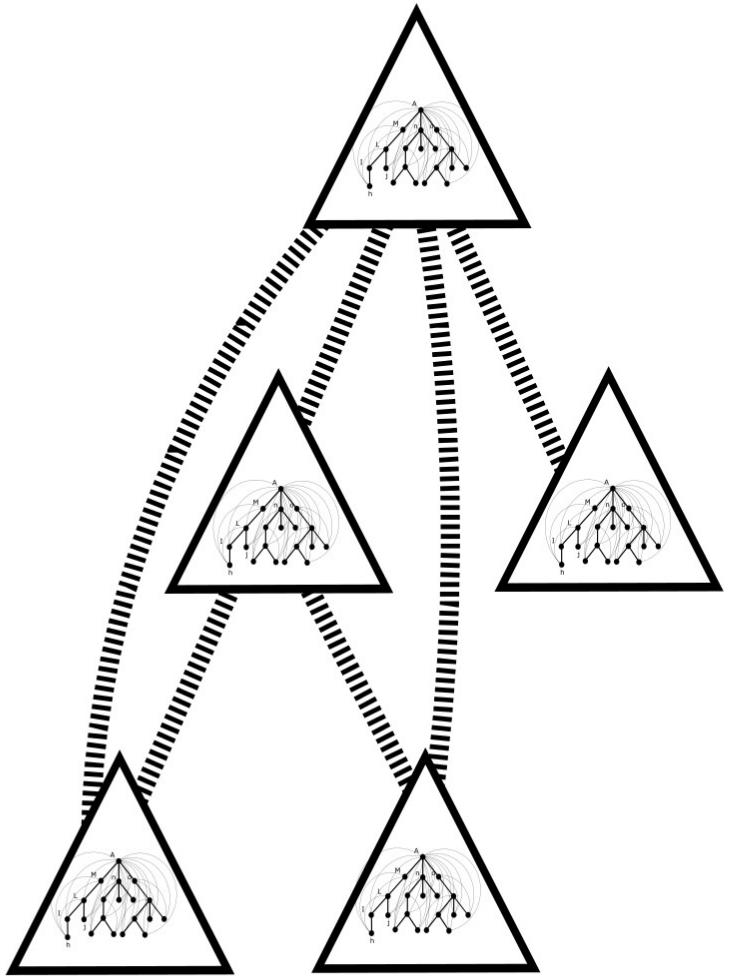
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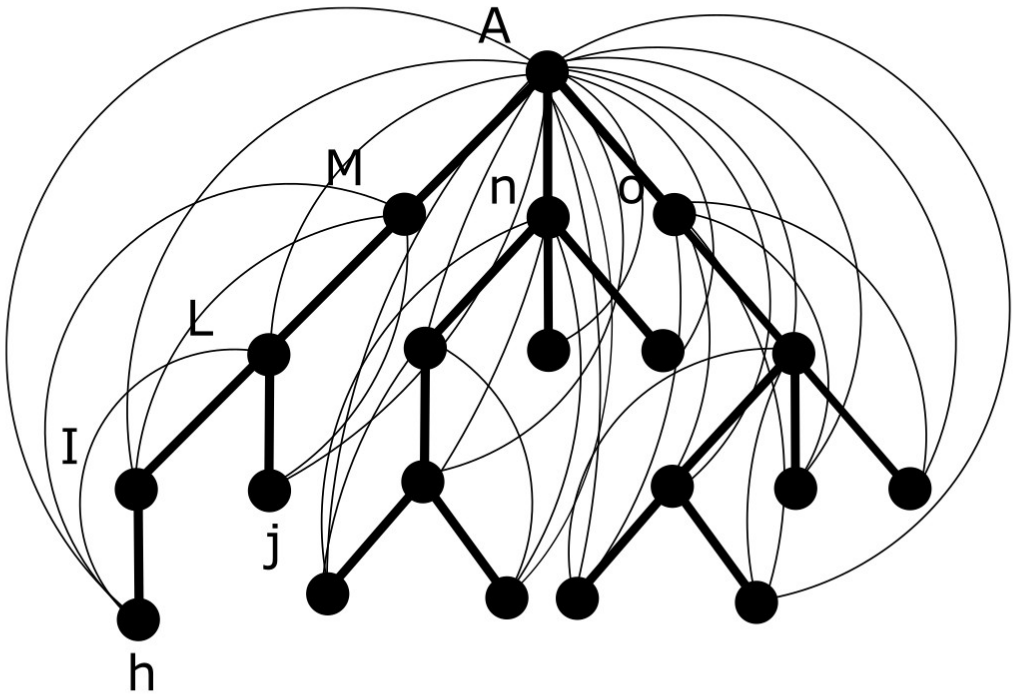
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Result 1: threshold graph → quasi-threshold → cograph

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cographs

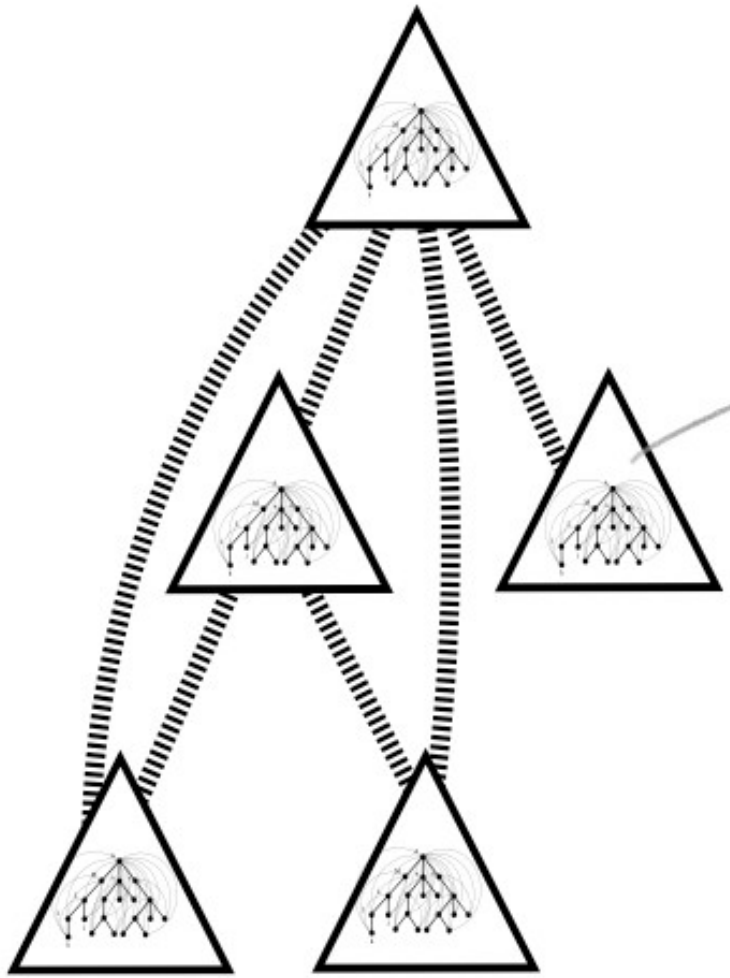


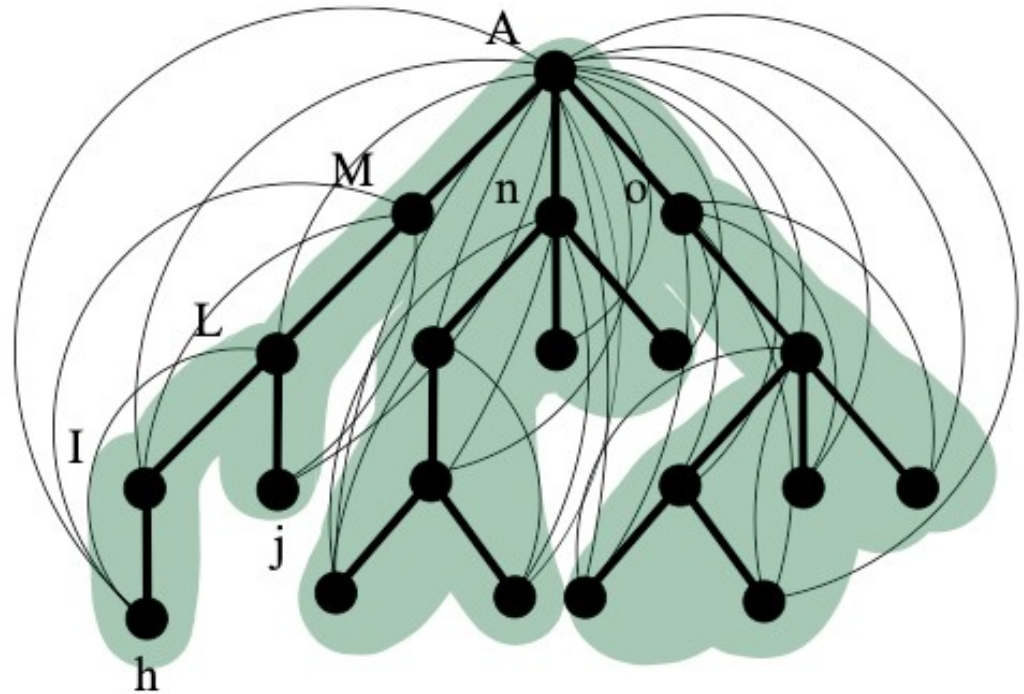


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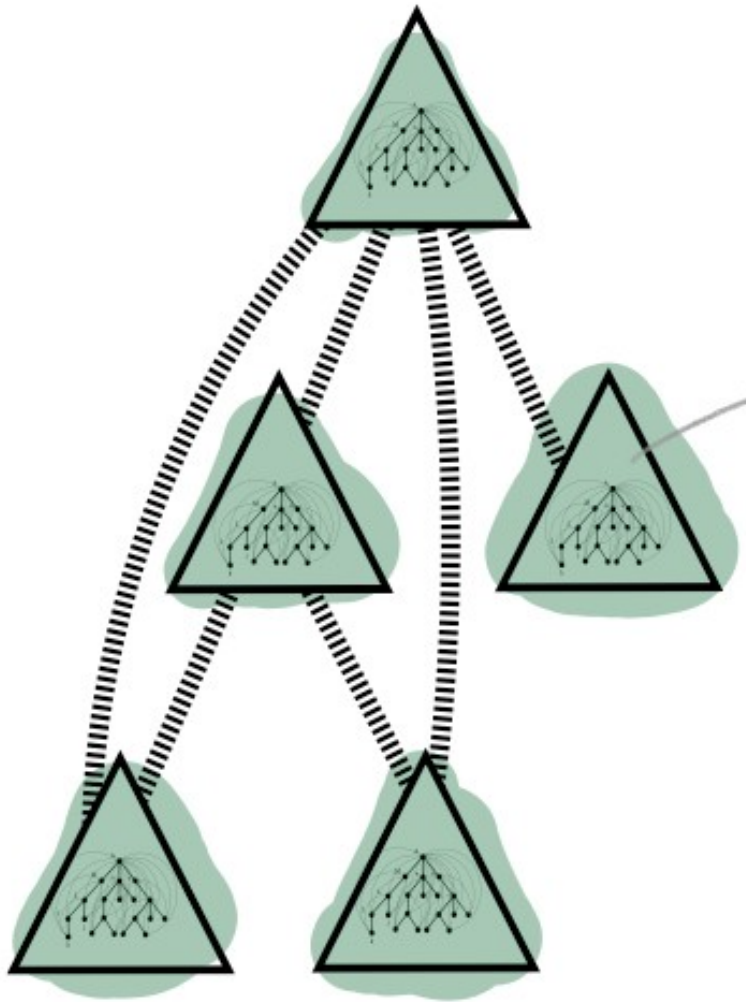


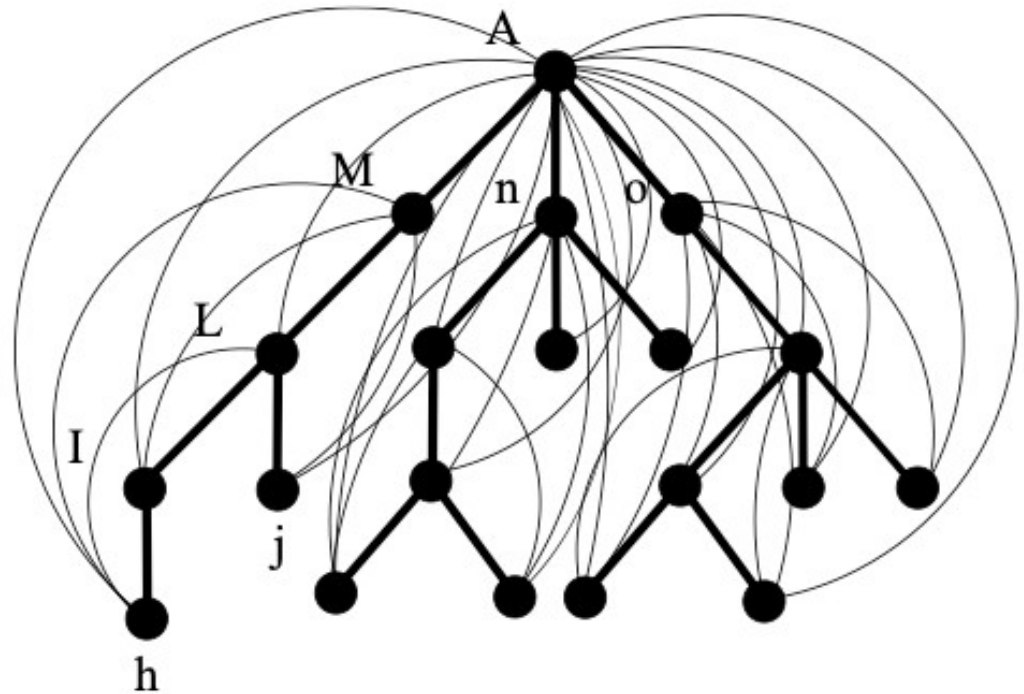


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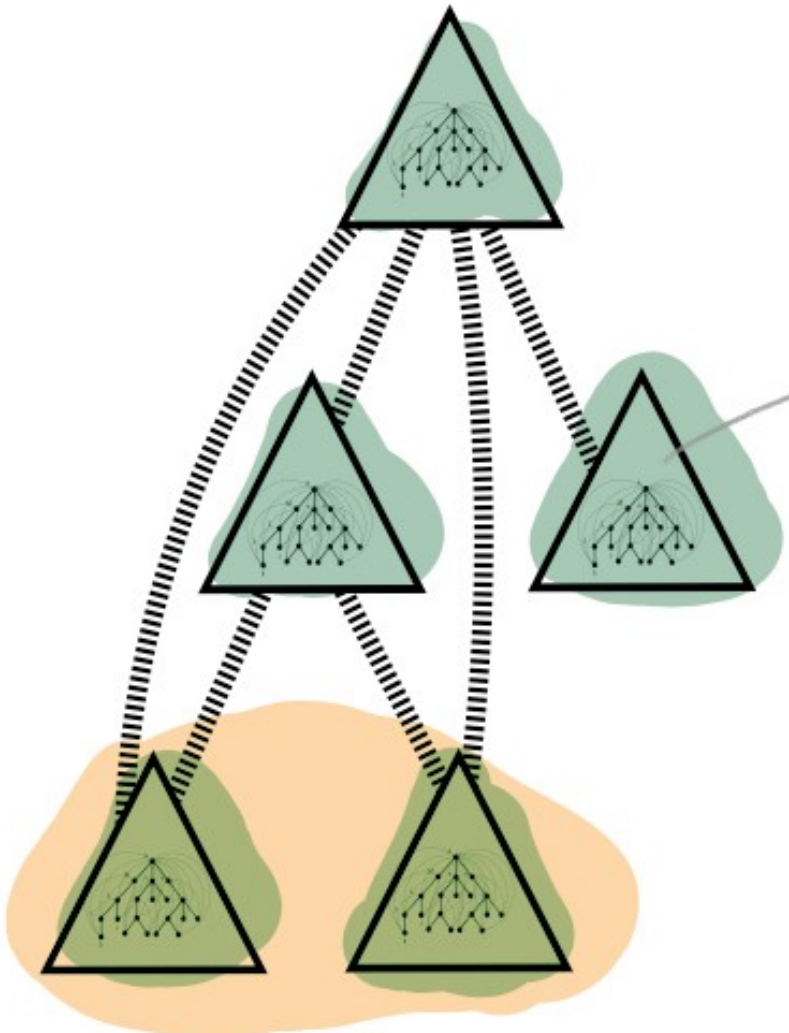


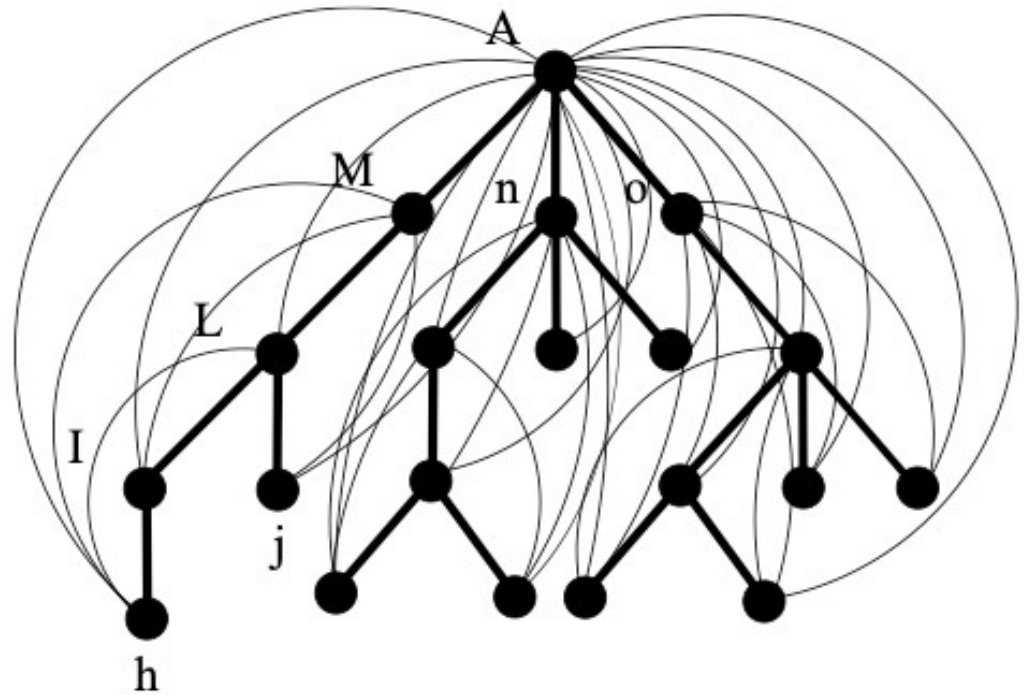


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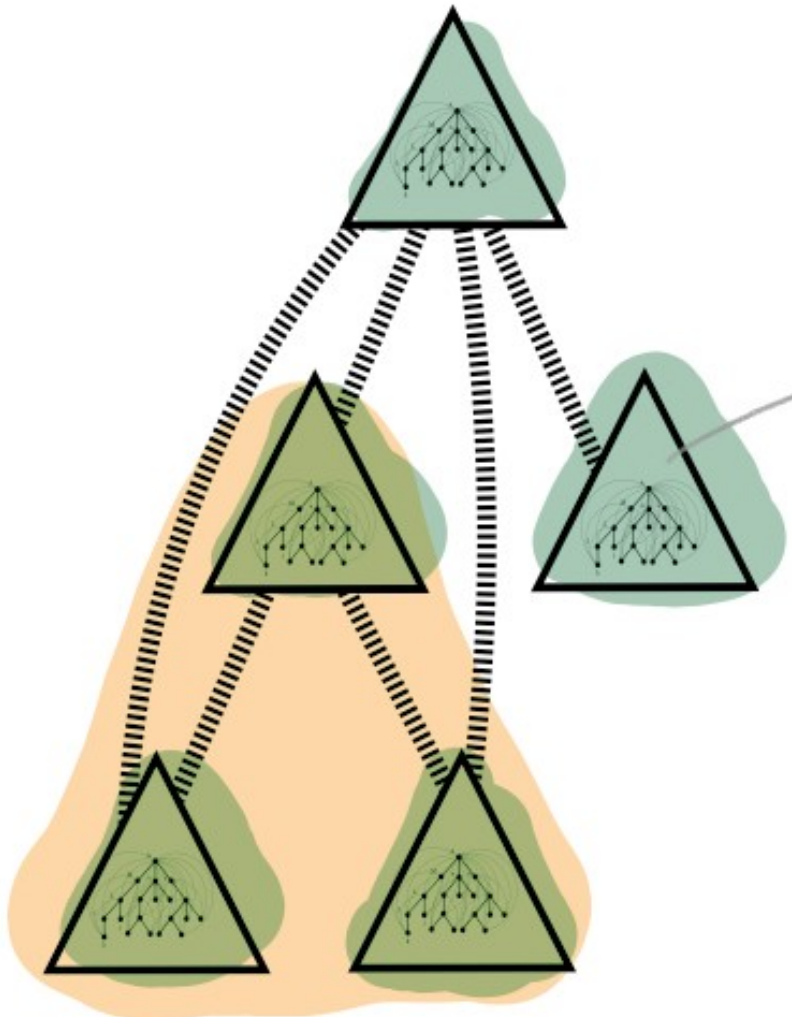


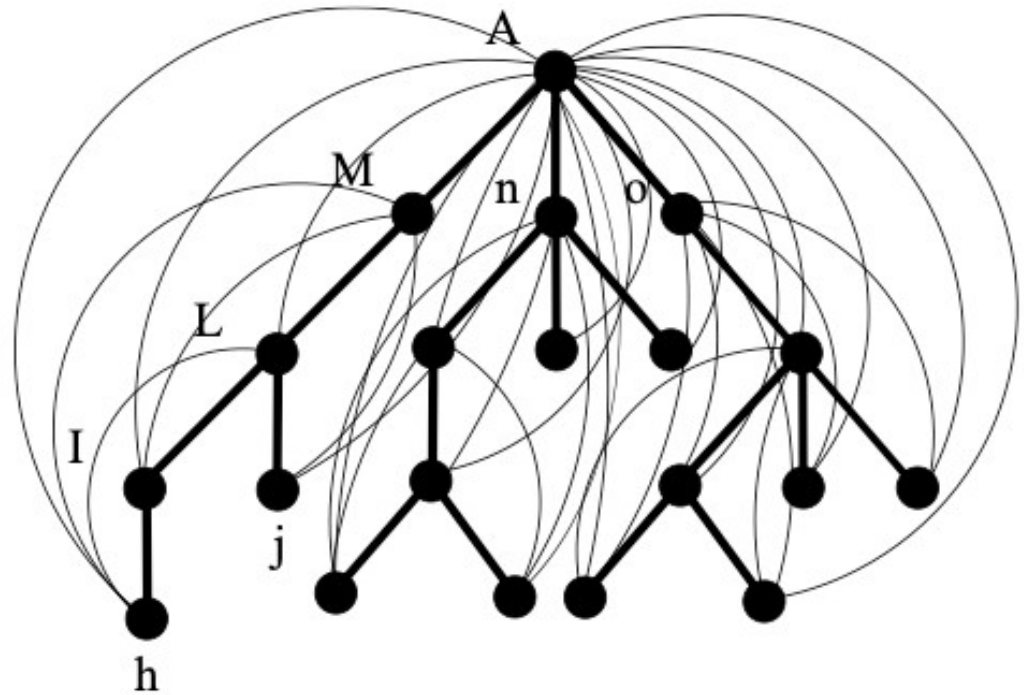


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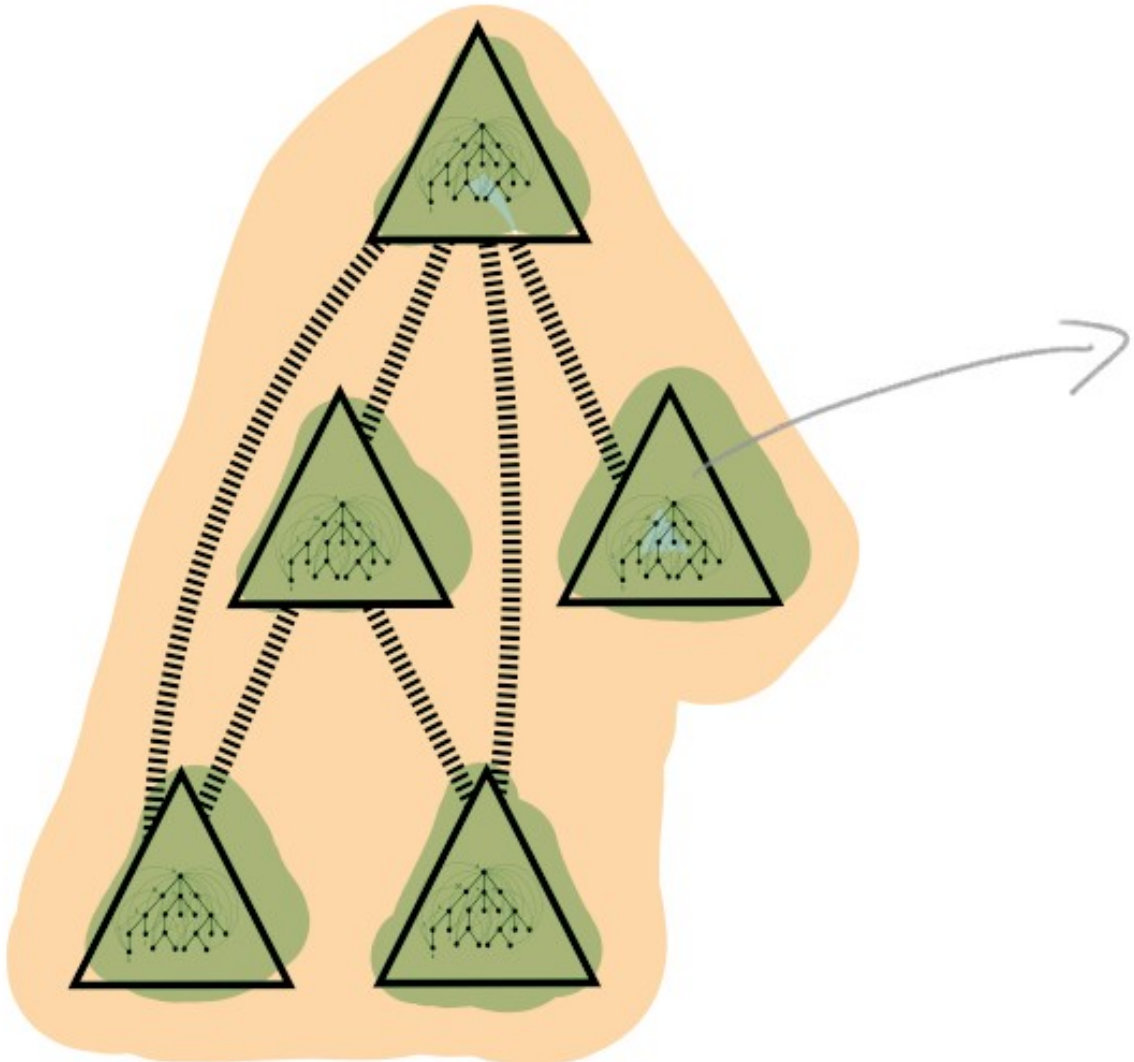


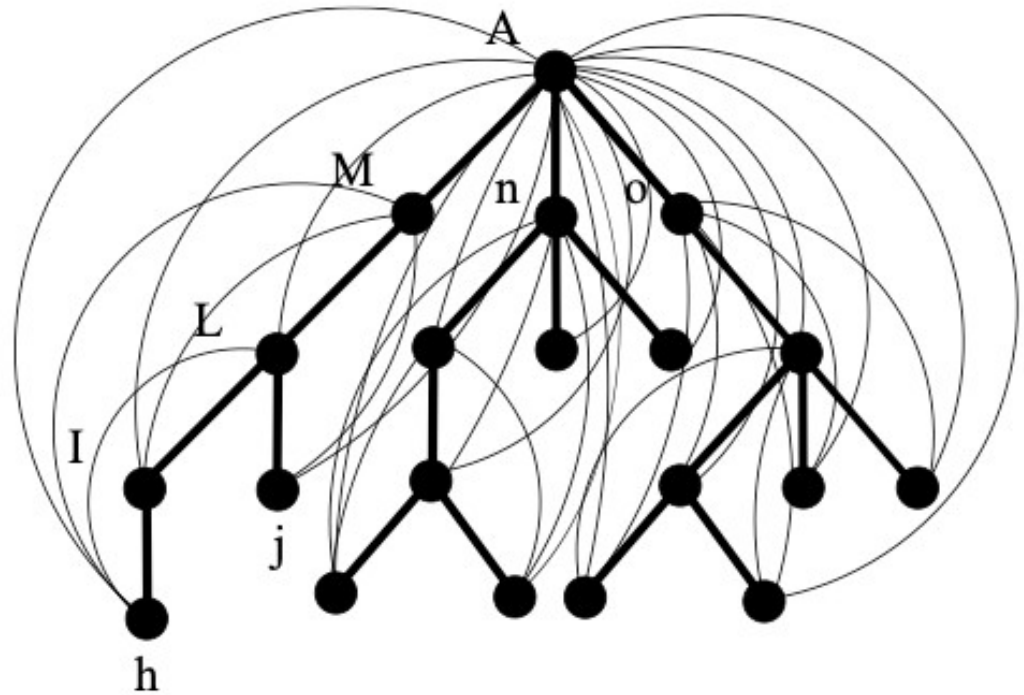


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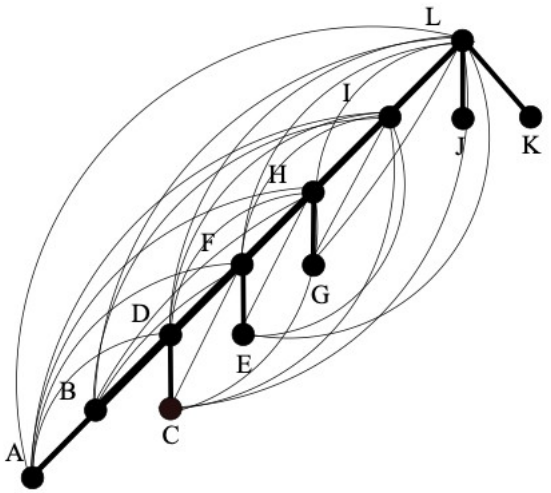
cographs

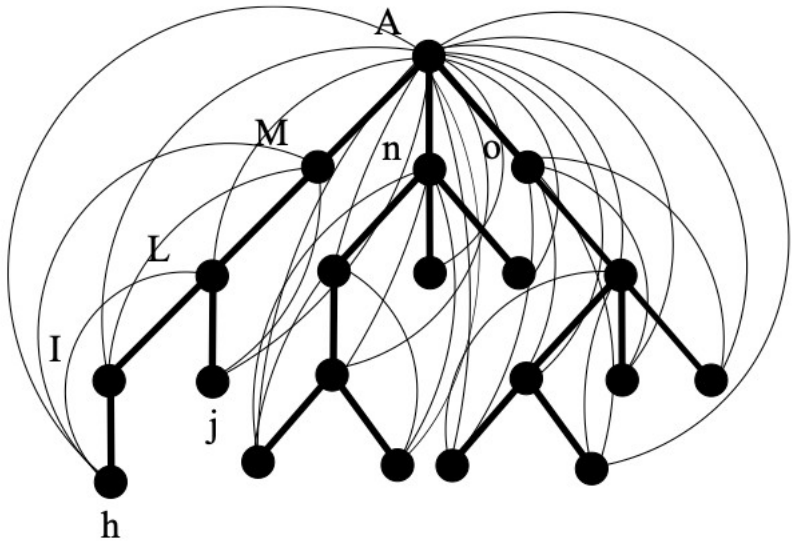


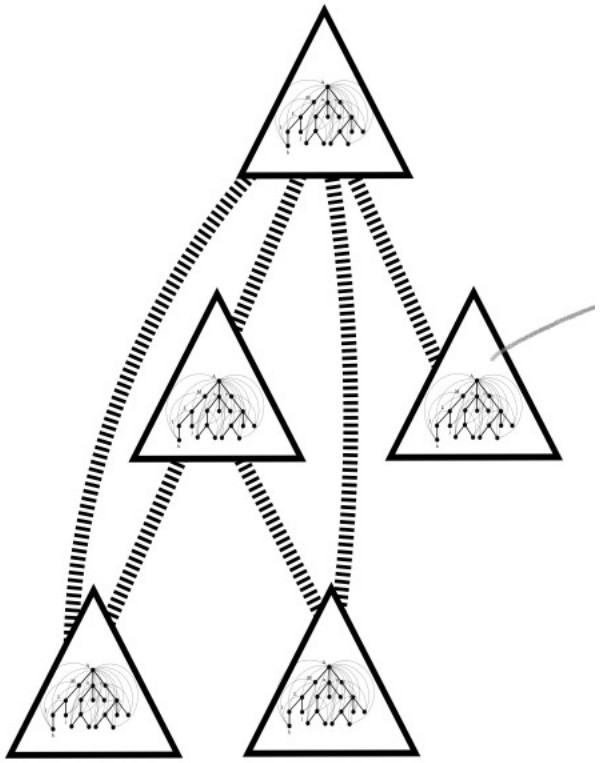


Summarize

- Contributes three new classes of global sync graphs.

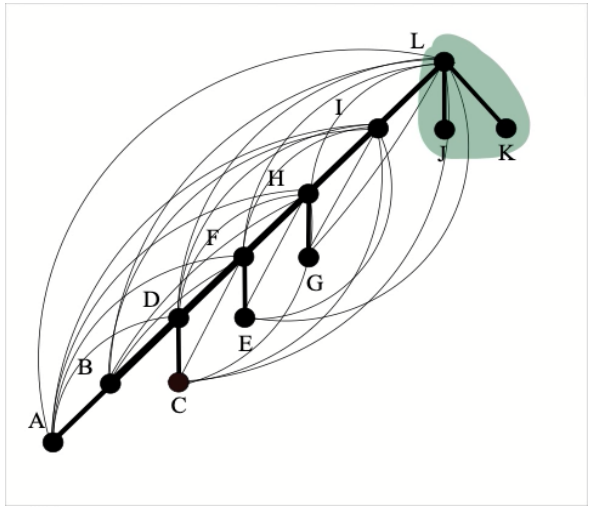


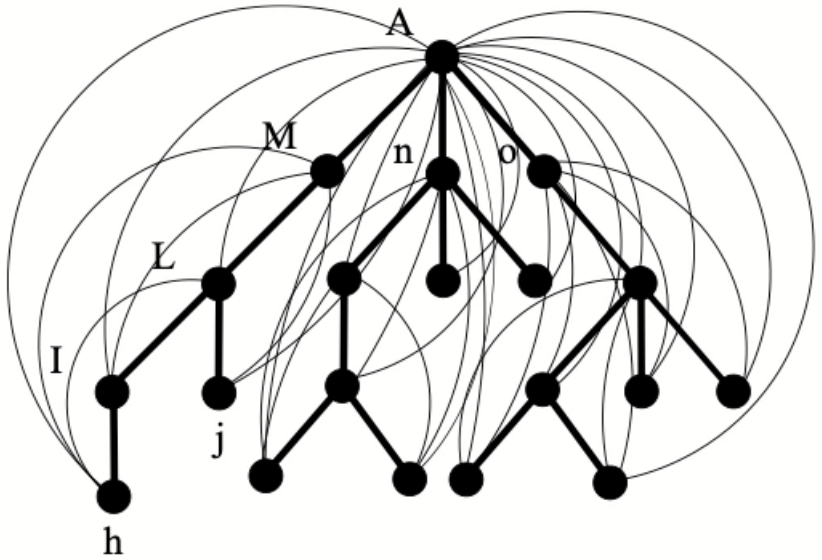


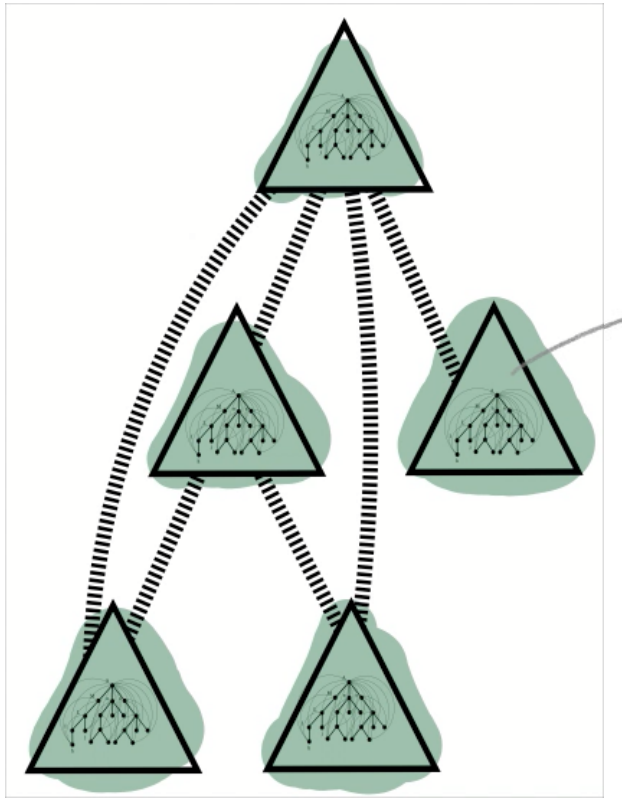


Summarize

- Contributes three new classes of global sync graphs.
- What we've learned from this: local to global propagation.

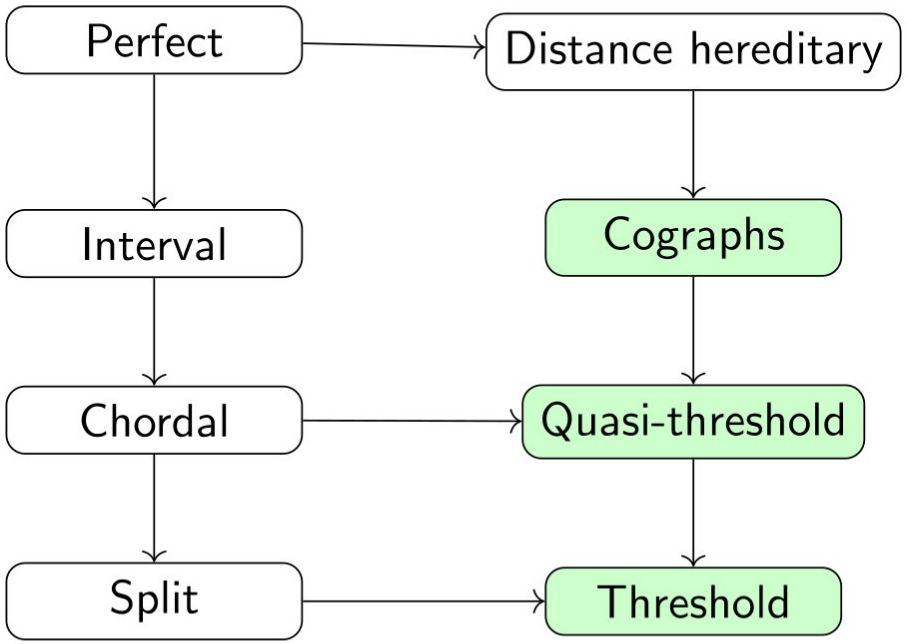






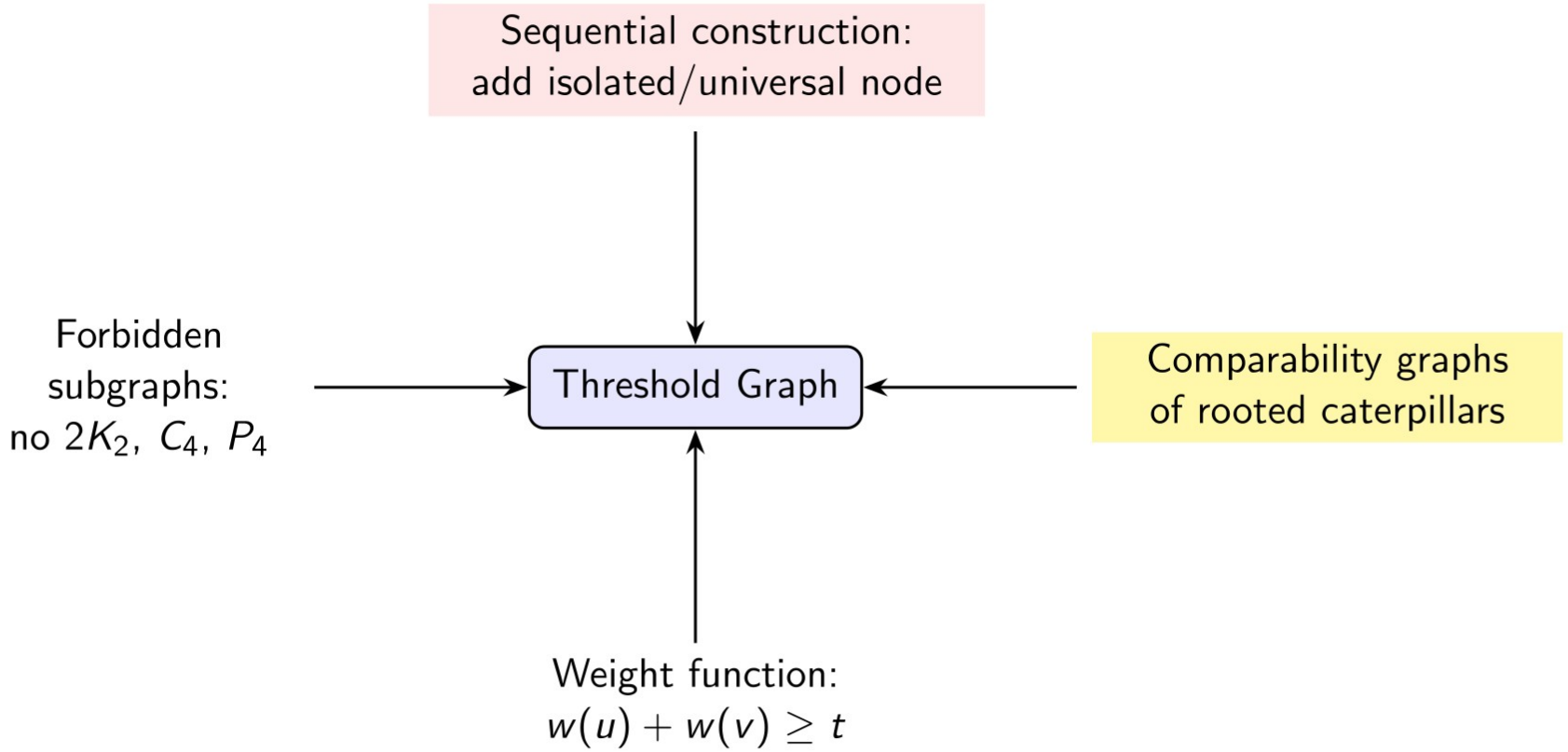
Outlook

- Extend the results: among all globally synchronizing graphs, how much of them can actually be explained by this propation mechanism?



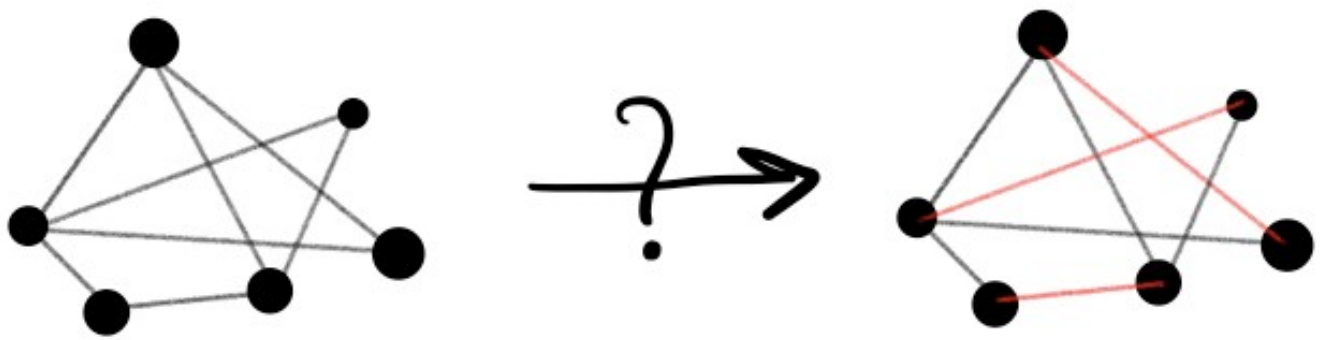
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$$\min_{\theta} \; - \sum_{i=1}^{n} \sum_{j=1}^{n} A_{ij} g\left(\cos(\theta_i - \theta_j)\right)$$

End..



Birds Synchronization @Rapperswil, 2022.03.14